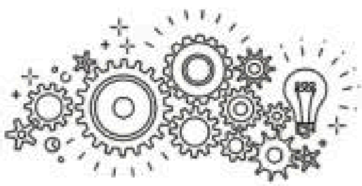
Anticancer agents I

Krisada Sakchaisri, Ph.D.

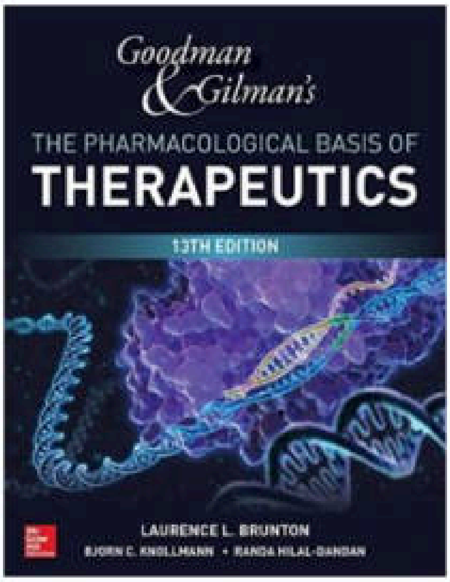
Learning Objectives

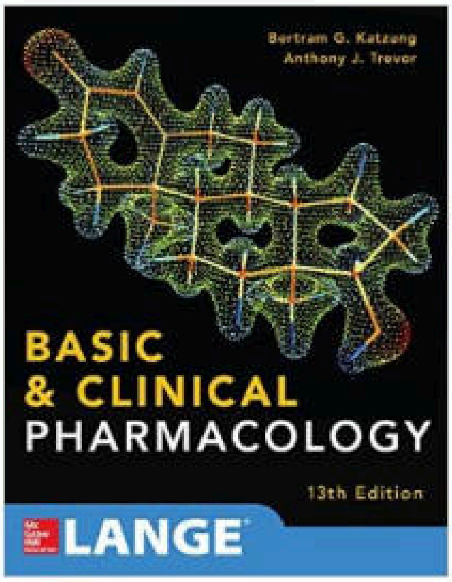


- Explain and compare the mechanism of actions of anticancer agents

- Explain general and specific adverse effects of anticancer agents

Textbooks





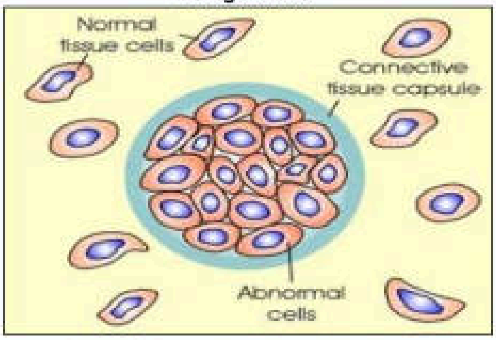
Terminology

- **Tumor or Neoplasm**
  - ☞ *An abnormal mass of tissue as a result of abnormal growth or division of cells*
  - ☞ ***Not all tumors are cancerous***
    - **Benign tumor (non-cancerous)**
      - ☞ *Cells in benign tumors do not spread to other parts of the body*
      - ☞ *Benign tumors are* designated by attaching the suffix ***-oma*** to the cell type from which the tumor arises
    - **Malignant tumor (cancerous)**
      - ☞ *Cells in these tumors can invade nearby tissues and spread to other parts of the body*

Cancer ➠ *Diseases in which abnormal cells divide without control and are able to invade other tissues*

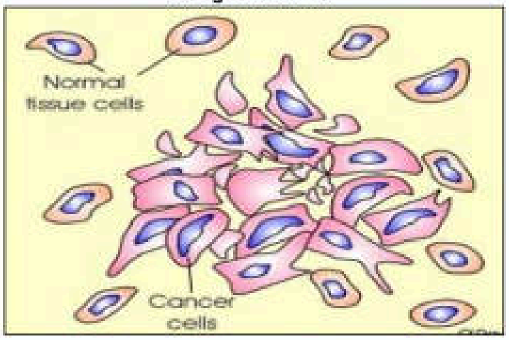
Characteristics of Benign and Malignant Tumors

Benign Tumor



- Well-differentiated cells that resemble cells in the tissue of origin
- Grows by expansion without invading the surrounding tissue
- No metastasis

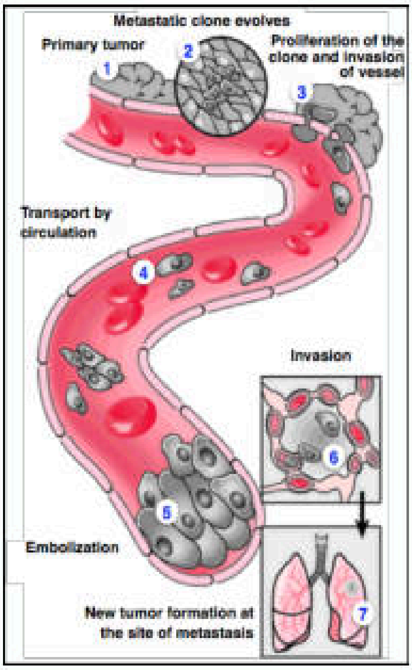
Malignant Tumor



- Anaplasia (reversion of differentiation)
- Atypical structure, little resemblance to cells in the tissue of origin
- Grows by invasion
- Metastasis

Canc Therapy & Oncol Int J 10(3): CTOIJ.MS.ID.555790 (2018)

Terminology

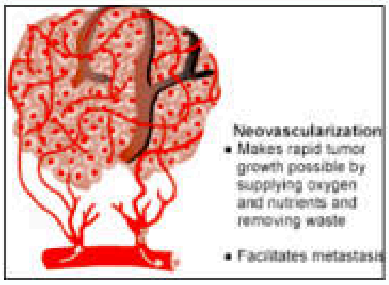


- **Metastasis**

  ☞ *Ability to invade adjacent normal tissues and break away from the primary tumor and travel through the blood or lymph to establish new tumors at a distant site*

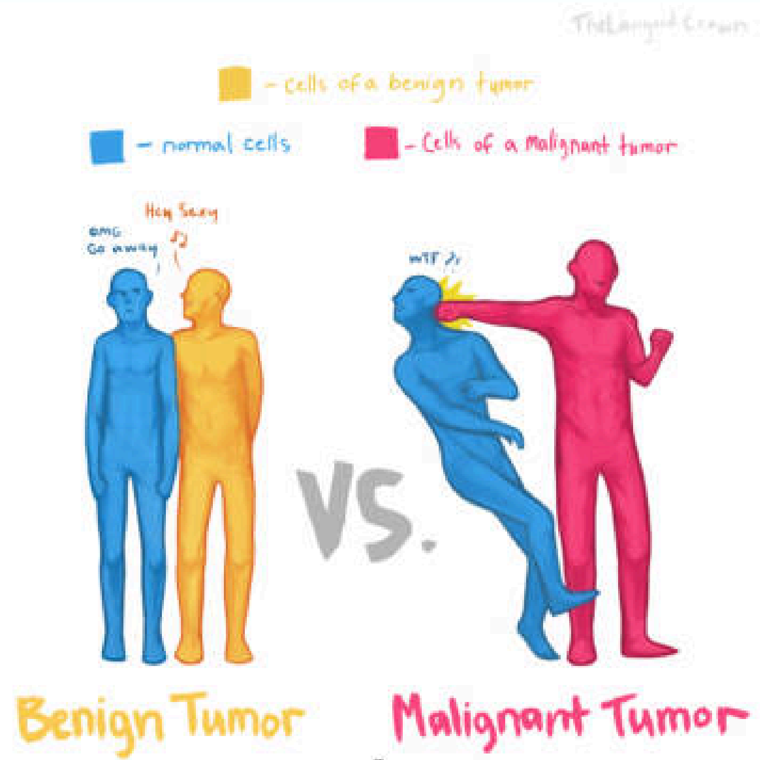
- **Angiogenesis**

  ☞ *Formation of new blood vessels*



http://medical-dictionary.thefreedictionary.com/metastasis

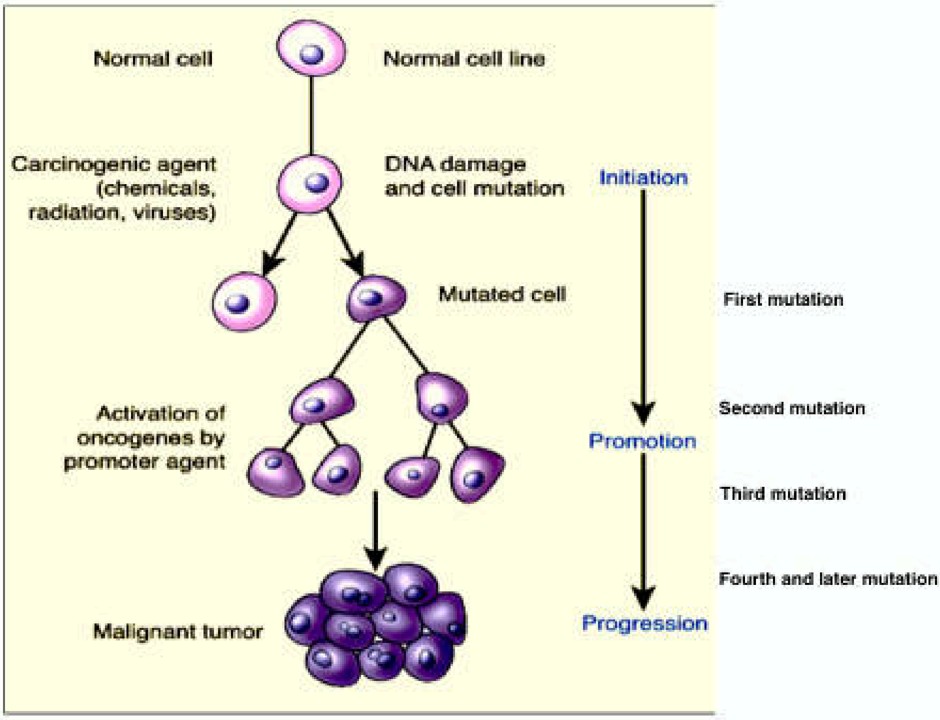
www.clinicaloptions.com



Thelanguidclown.deviantart

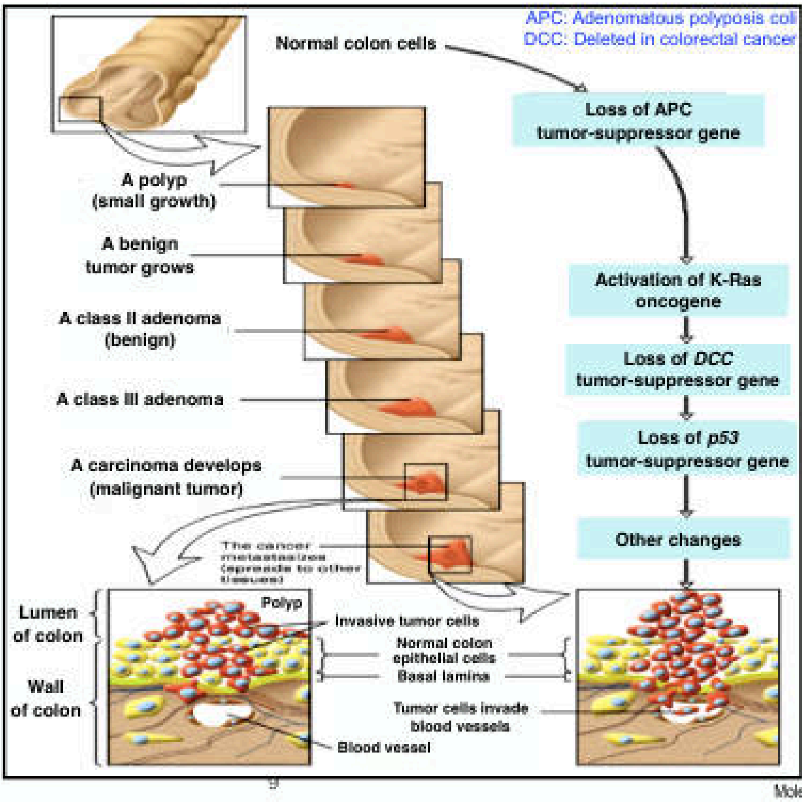
Carcinogenesis:

A Multistep Process



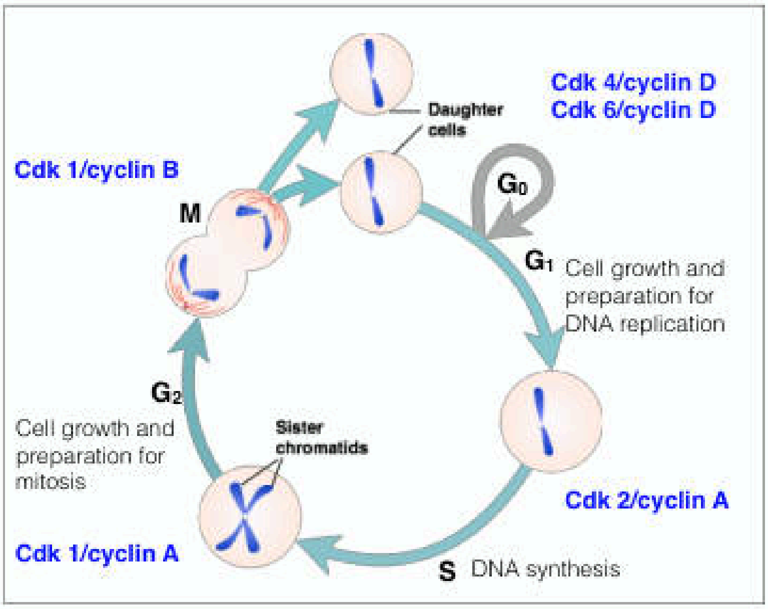
Pathophysiology: Concepts of altered health states, 8th Ed

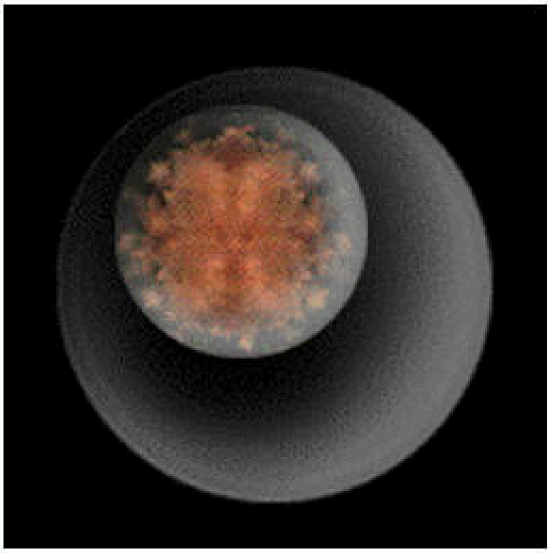
The Development and Metastasis of Human Colorectal Cancer and Its Genetic Basis



Molecular Cell Biology, 6th edition

The Eukaryotic Cell Cycle





Modalities of Cancer Treatment

Surgery

Radiation therapy

Systemic therapy

- Chemotherapy
- Targeted therapy
- Immunotherapy

The Systemic Anticancer Therapy

Alkylating Agents

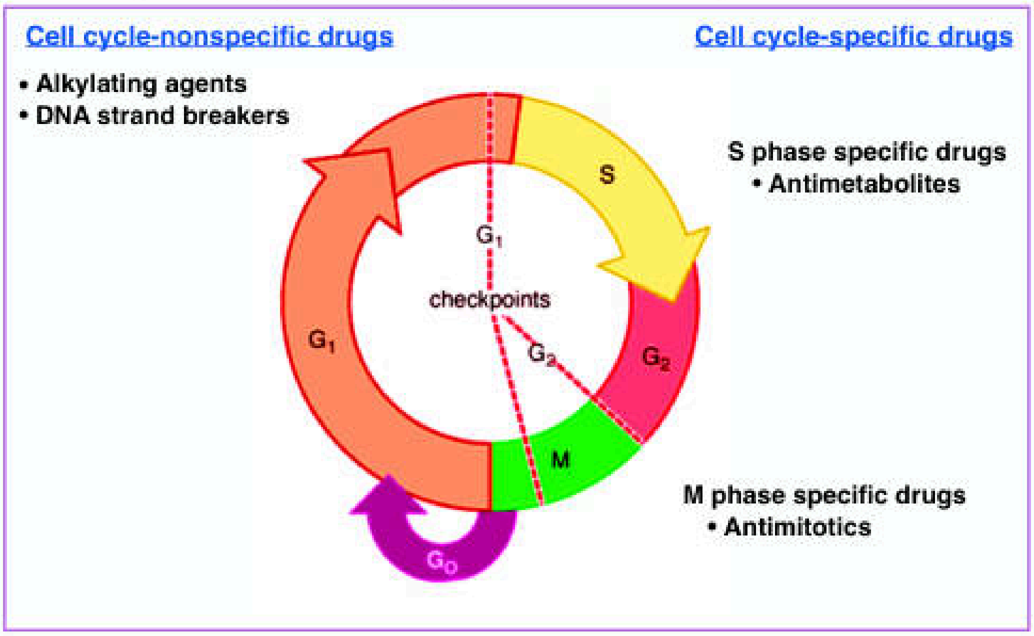
Antimetabolites

Natural Product Cancer Chemotherapeutic Drugs

Antitumor antibiotics

Targeted Therapy

Cell Cycle Specificity of Antineoplastic Agents



Modified from *Goodman and Gilman's: The Pharmacological Basis of Therapeutics, 12th Ed*

Cell cycle-nonspecific (CCNS) drugs

- These drugs have a **linear dose-response curve**

  "The greater amount of drug, the greater fraction of cells killed"

- Drugs of this nature are often given as a single bolus injection

Cell cycle-specific (CCS)drugs

- These drugs act only at particular phases of the cell cycle and have a plateau with respect to cell killing ability

  "The amount of cell kill will not increase as drug dosage increases"

- Cell cycle specific drugs are administered in minimal concentration via continuous dosing methods

Alkylating Agents

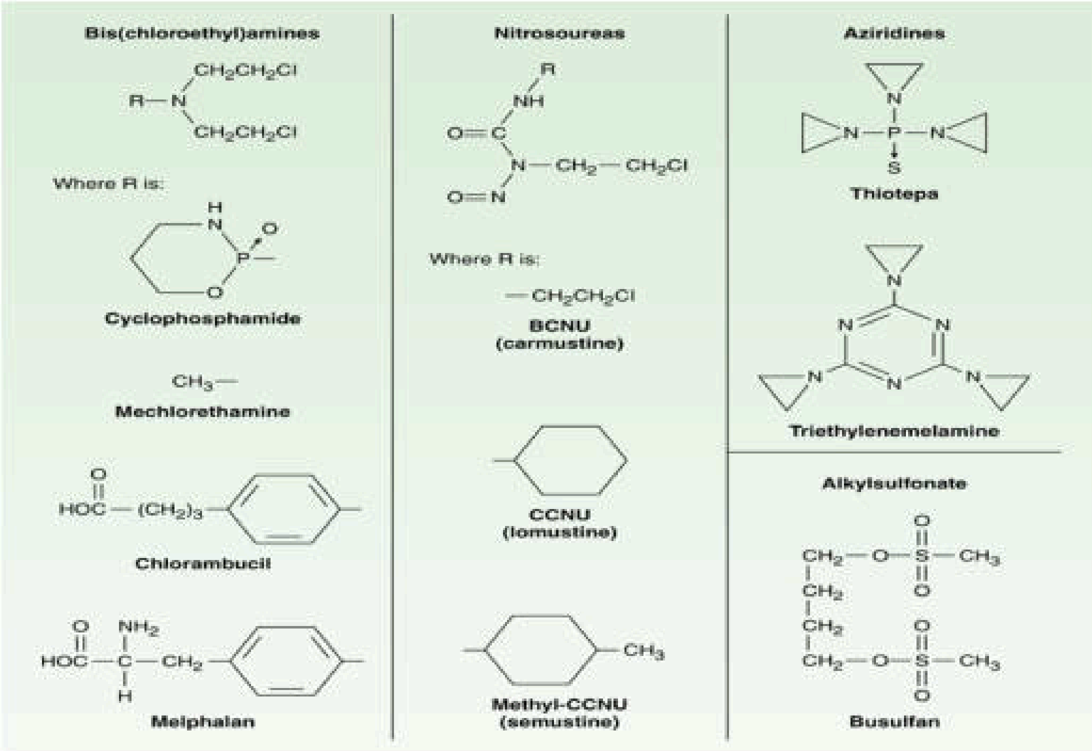
Alkylating Agents

- Directly damage the DNA
  - ➠ preventing the cancer cells from multiplying
- Classified in several different groups according to the moiety in their structures

Type of Agent	Name
Nitrogen mustards	Cyclophosphamide Mechlorethamine Ifosfamide Melphalan Chlorambucil
Ethyleneimines/ Aziridines	Thiotepa Triethylenemelamine Atretamine
Triazines	Dacarbazine Temozolomide
Methylhydrazine derivatives	Procarbazine

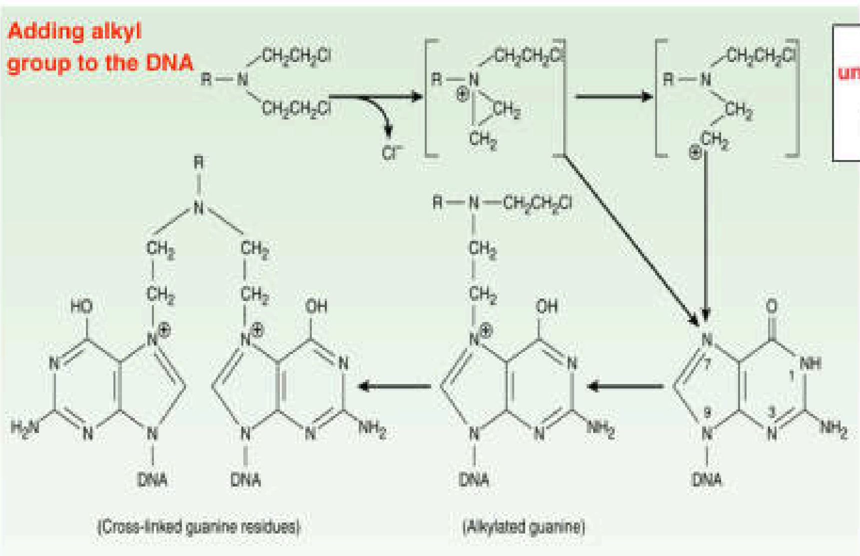
Type of Agent	Name
Nitrosoureas	Carmustine (BCNU) Lomustine (CCNU) Semustine Bendamustine Streptozotocin
Alkyl sulfonate	Busulfan
Platinum complex	Cisplatin Carboplatin Oxaliplatin

Structure of Major Classes of Alkylating Agents



Katzung BG, Masters SB, Trevor AJ: Basic & Clinical Pharmacology

Alkylating Agents: Mechanism of Action



Nucleophilic attack of unstable aziridine ring by electron donor

(–SH of protein, –N– of protein or DNA base, =O of DNA base or phosphate)

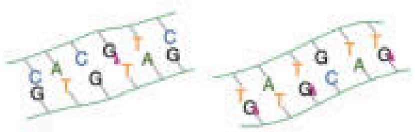
- Alkylating agents attack DNA to a preferentially generate **N7-alkylguanine**
- Other bases are also alkylated with lesser degree, including N1 and N3 of adenine, N3 of cytosine, and O6 of guanine

Katzung BG, Masters SB, Trevor AJ: Basic & Clinical Pharmacology

Alkylating Agents: Mechanism of Action

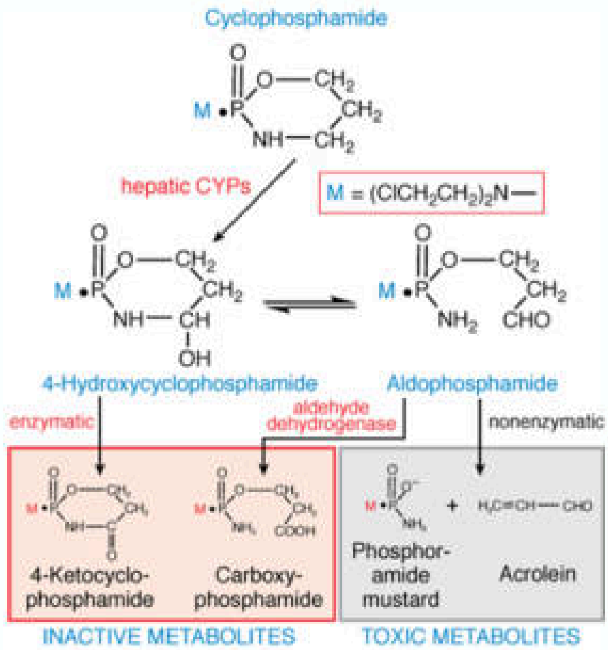
- The modified guanine can **mispair** with thymine residues during DNA synthesis
- Alkylation of N7 creates lability in the imidazole ring, causing **strand breakage**
- **Cross-linking** of two nucleic acid chains or the linking of a nucleic acid to a protein

☞ Major **disruption** in nucleic acid function



Prevent replication or transcription

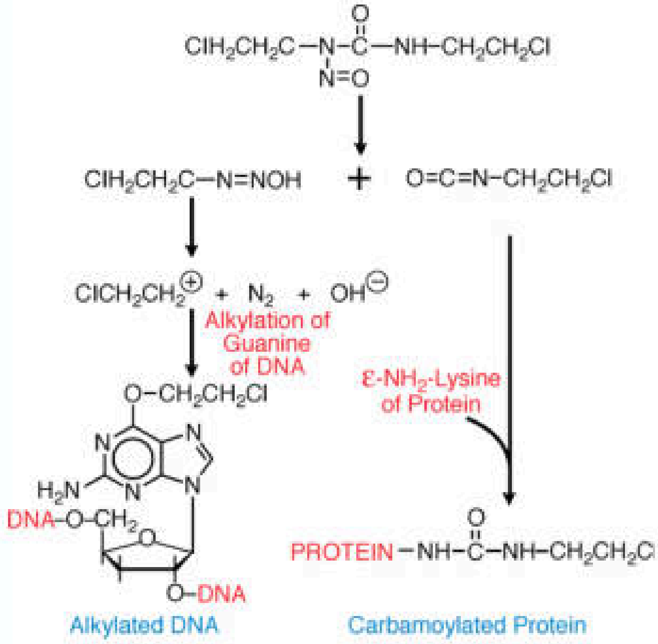
Cyclophosphamide



- One of the most widely used alkylating agents
- High oral bioavailability (clinical efficacy: oral = IV)
- Liver appears to be protected through the enzymatic formation of the inactive metabolites
- Hemorrhagic cystitis:
  - Bladder damage is due to ***acrolein***, which has a direct toxic effect on the mucosa
  - Mesna detoxifies acrolein and, thereby, protects bladder toxicity

Brunton LL, Chabner BA, Knollmann BC: Goodman & Gilman's Th Pharmacological Basis of Therapeutics

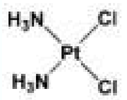
Nitrosoureas



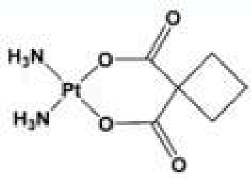
- Highly lipid-soluble (able to cross BBB)
- Alkylation
  - N7 guanine (major)
  - O6 guanine (critical for cytotoxicity)
- Non-cross-resistant with other alkylating agents

Brunton LL, Chabner BA, Knollmann BC: Goodman & Gilman's Th Pharmacological Basis of Therapeutics

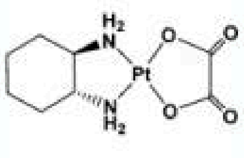
Platinum Analogs



cisplatin

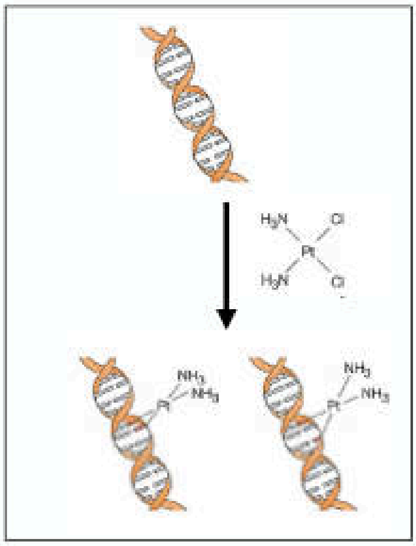


carboplatin

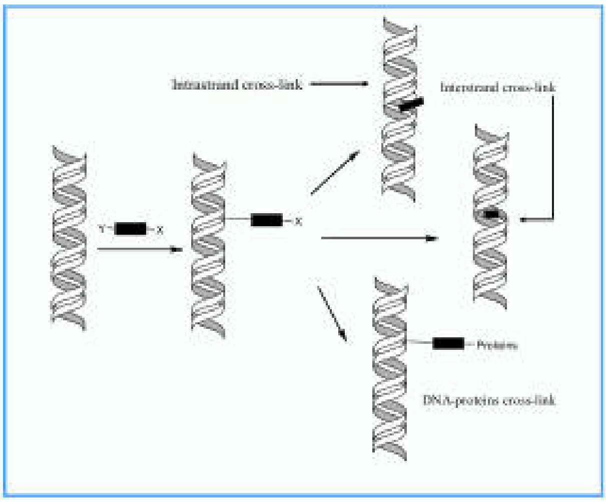


oxaliplatin

- **Cisplatin**
  - *Antitumor activity in a broad range of solid tumor*
  - *Extensively cleared by the kidneys*
  - ***Nephrotoxicity** (required vigorous IV hydration)*
- **Carboplatin**
  - *2nd-generation platinum analog*
  - *Less renal and GI toxicity*
  - *Main dose-limiting toxicity is myelosuppression*
- **Oxaliplatin**
  - *3rd-generation platinum analog*
  - *Not cross-resistant to cisplatin and carboplatin (on the basis of mismatch repair defect)*
  - *Main dose-limiting toxicity is peripheral sensory neuropathy*



Alkylating Agents



- ✦ Attaching alkyl groups to DNA bases, leading to DNA fragmentation
- ✦ Causing intra- and inter-strand DNA cross-links
- ✦ Inducing mispairing of nucleotides, leading to mutations
- ✦ Exerting cytotoxicity in all phases of the cell cycle

Adverse Effects

✦ Exerting on rapidly proliferating cells

: **skin**—hair loss

: **GI tract**—diarrhea, nausea, and vomiting

: **bone marrow**—anemia causing fatigue/leukopenia causing infections/thrombocytopenia causing bleeding)

Alkylating Agent	Mechanism of Action	Clinical Applications	Toxicity
Mechlorethamine	Form DNA cross-link ➡ inhibition of DNA synthesis and function	Hodgkin's and non-Hodgkin's lymphoma	• Nausea and vomiting • Myelosuppression • Hemorrhagic cystitis (cyclophosphamide) • Skin pigmentation (busulfan) • Pulmonary fibrosis (busulfan) • Adrenal insufficiency (busulfan)
Chlorambucil	Same as above	Chronic lymphocytic leukemia (CLL) and non-Hodgkin's lymphoma	
Cyclophosphamide	Same as above	Breast cancer, ovarian cancer, non-Hodgkin's lymphoma, CLL, soft tissue sarcoma, Wilm's tumor, neuroblastoma, rhabdomyo sarcoma	
Melphalan	Same as above	Multiple myeloma, breast cancer, ovarian cancer	
Thiotepa	Same as above	Breast cancer, ovarian cancer, superficial bladder cancer	
Busulfan	Same as above	Chronic myeloid leukemia (CML)	
Lomustine	Same as above	Brain cancer	• Nausea and vomiting • Myelosuppression • Interstitial lung disease (rarely) • Interstitial nephritis (rarely)
Carmustine	Same as above	Brain cancer, Hodgkin's and non-Hodgkin's lymphoma	

Basic and Clinical Pharmacology, *12th Ed*

Alkylating Agent	Mechanism of Action	Clinical Applications	Toxicity
Temozolomide	Methylates DNA and inhibits DNA synthesis and function	Brain cancer, melanoma	• Nausea and vomiting • Myelosuppression • Photosensitivity
Procarbazine		Hodgkin's and non-Hodgkin's lymphoma, brain tumors	• Nausea and vomiting • Myelosuppression • Hypersensitivity reactions
Dacarbazine		Hogkin's lymphoma, melanoma, soft tissue sarcoma	• Nausea and vomiting • Myelosupprssion • CNS toxicity
Melphalan		Multiple myeloma, breast cancer, ovarian cancer	• Nausea and vomiting • Myelosuppression
Cisplatin	Form intrastrand and interstrand DNA cross-link, binding to nuclear and cytoplasmic proteins	NSCLC, small cell lung cancer (SCLC), breast cancer, bladder cancer, cholangiocarcinoma, gastroesophageal cancer, head and neck cancer, ovarian cancer	• Nausea and vomiting • Myelosuppression • Nephrotoxicity • Peripheral sensory neuropathy • Ototoxicity
Carboplatin		NSCLC, SCLC, breast cancer, bladder cancer, head and neck cancer, ovarian cancer	• Nausea and vomiting • Myelosuppression • Nephrotoxicity

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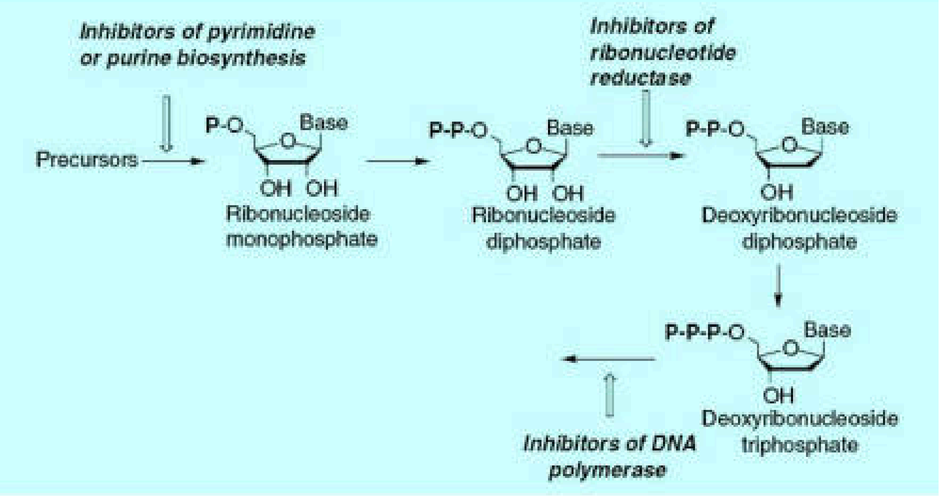
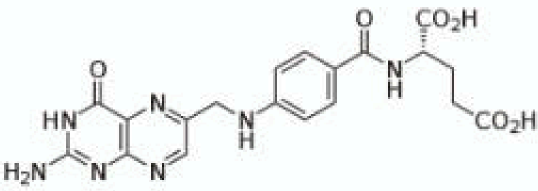
Antimetabolites

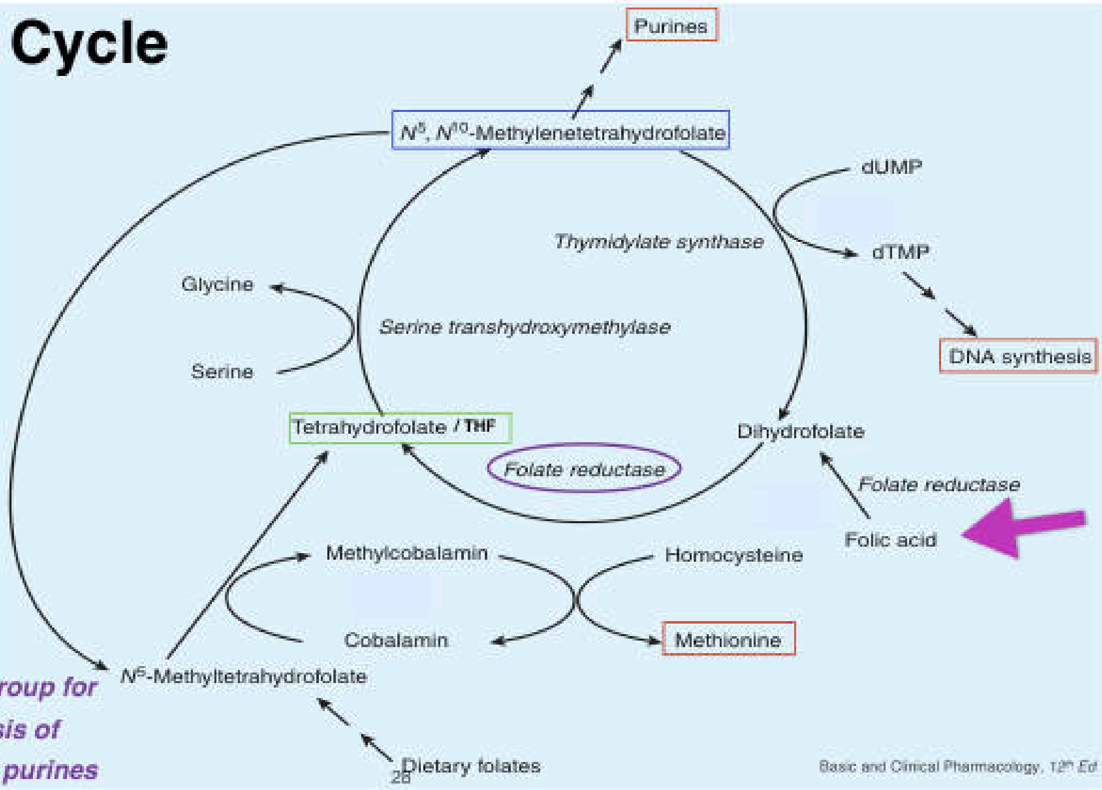
- **Antifolates**
- **Ribonucleotide reductase inhibitor**
- **Purine & Pyrimidine analogs**



Antifolates

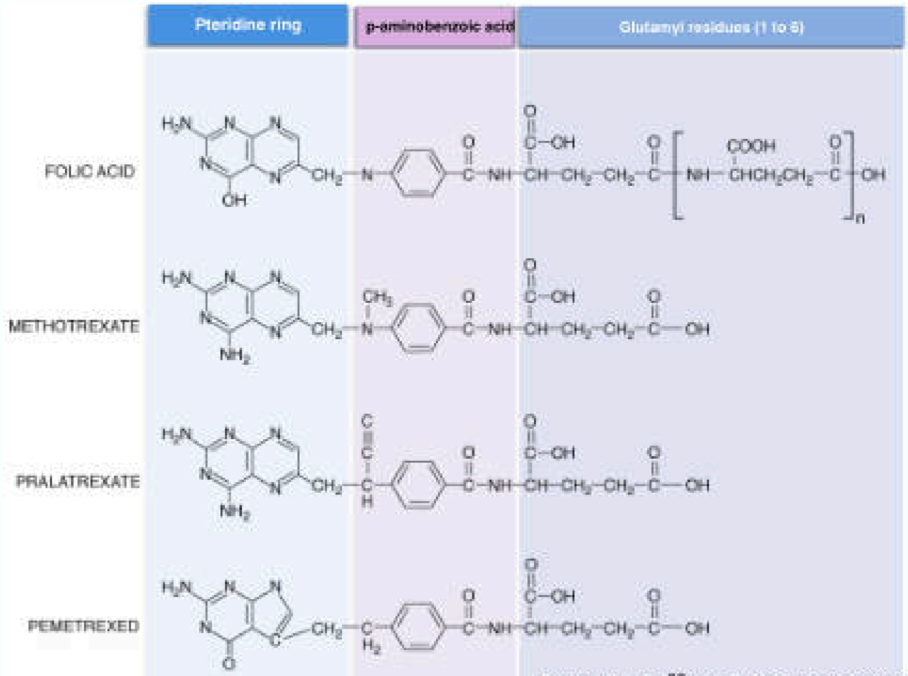


The Folate Cycle



THF provides methyl group for the de novo synthesis of methionine, dTMP, and purines

Folic Acid and Its Analog

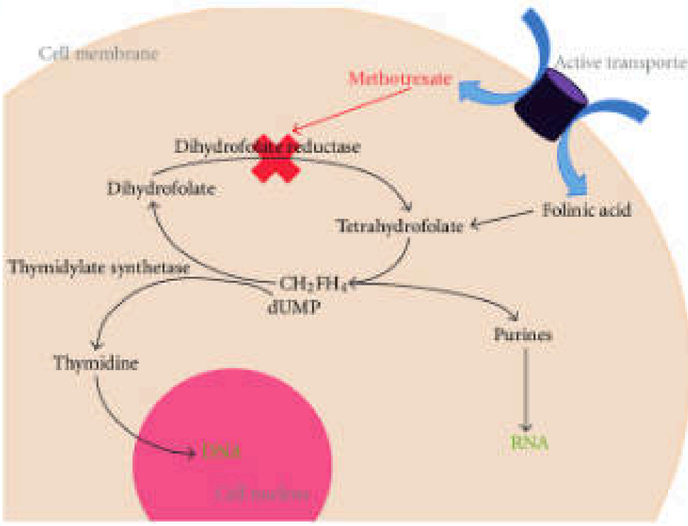


Folate antagonists kill cells during the S phase and are most effective when cells are proliferating rapidly

Brunton LL, Chabner BA, Knollmann BC: Goodman & Gilman's Th Pharmacological Basis of Therapeutics

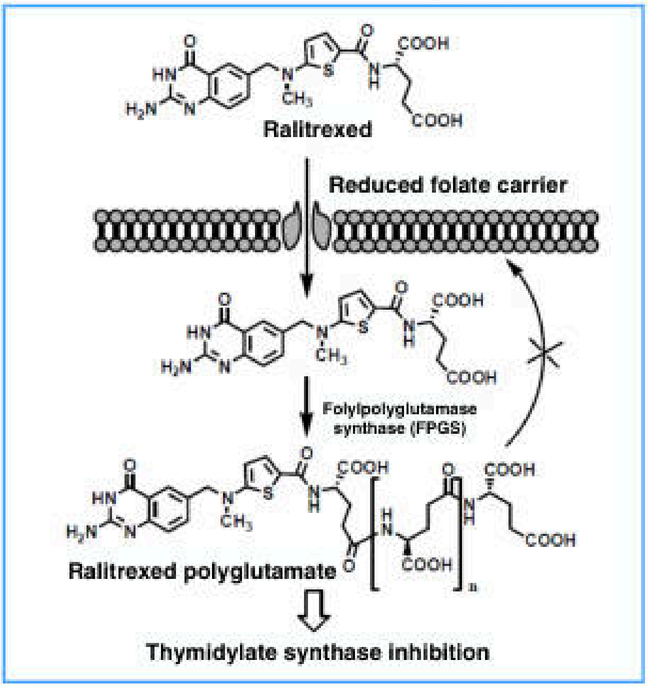
Mechnism of Action: Folate Analogs

- **Inhibits dihydrofolate reductase (DHFR)**
  - *Inhibition of the synthesis of THF*
  - *Interference with the formation of DNA, RNA, and key cellular proteins*



ISRN Obstet Gynecol. 2012;2012:637094

Role of Polyglutamate on The Antifolates



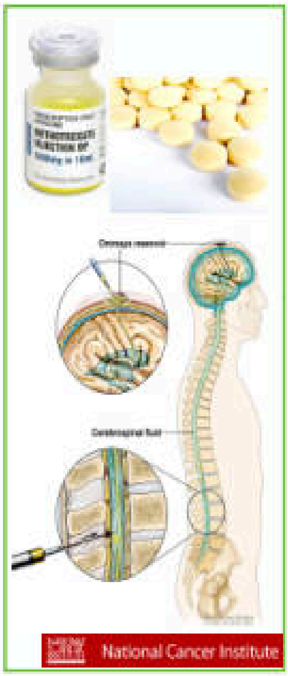
- **Polyglutamate metabolite (MTX polyglutamate)**
  - Critically important for the therapeutic action
  - Selectively retained within cancer cells
  - Increased inhibitory effects on enzymes involved in de novo synthesis of purine nucleotide and thymidylate

Methotrexate (MTX)

- **Administration**
  - *Intravenous, intrathecal, or oral route*
  - *Oral bioavailability is saturable (25 mg/m²)*
- **Elimination**
  - *Renal excretion (90% unchanged) is the main route of elimination*
  - *Dosage must be reduced in proportion to CrCl*
- **Drug Interaction**
  - *Caution when concomitantly using agents that inhibit renal excretion of MTX*

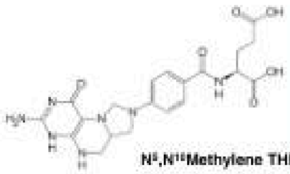
  aspirin, NSAIDs, penicillin, and cephalosporins

- **Adverse Effects**
  - *Diarrhea*
  - *Hand-foot syndrome*
  - *Myelosuppression*
  - *Nausea and vomiting*

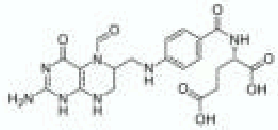


Leucovorin Rescue

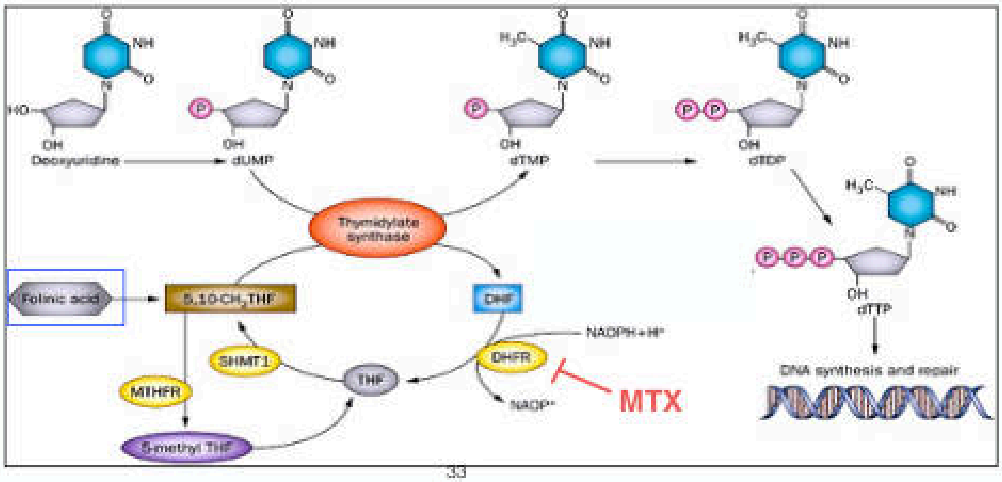
- The biological effects of MTX can be reversed by leucovorin *(folinic acid or 5-formyltetrahydrofolate)*
- Leucovorin rescue is used in conjunction with **high-dose MTX** therapy to rescue normal cells from undue toxicity



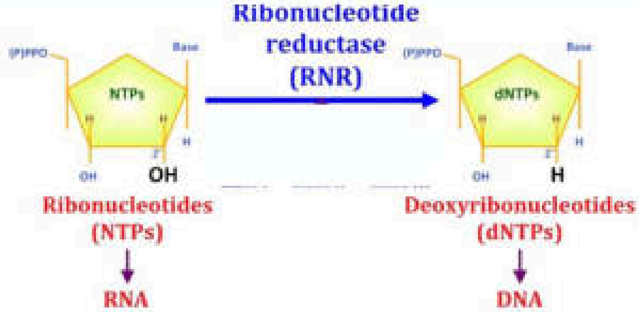
N^5,N^{10}Methylene THF



Leucovorin

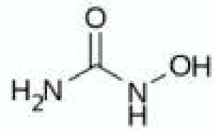


Ribonucleotide Reductase Inhibitor



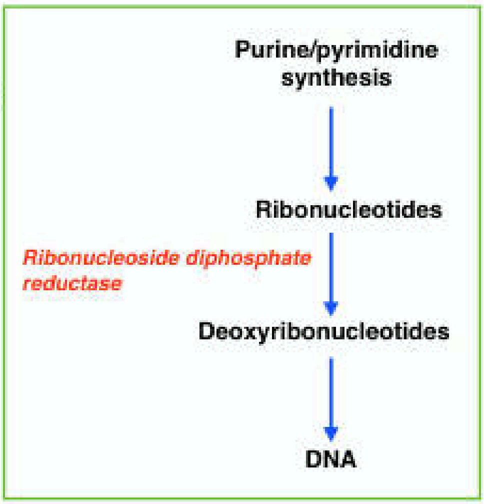
Front Cell Infect Microbiol. 2014 Apr 28;4:52

Hydroxyurea (HU)

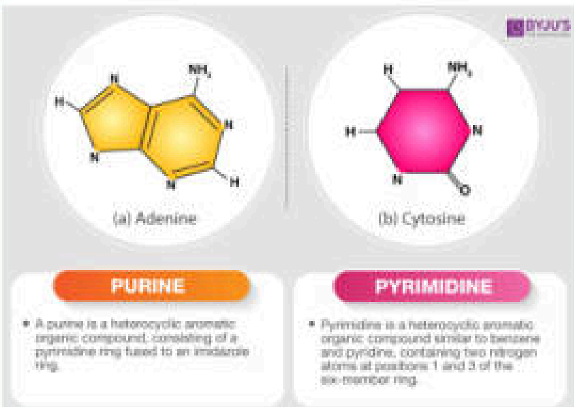


- Inhibits the enzyme **ribonucleoside diphosphate reductase**, which catalyzes the reductive conversion of ribonucleotides to deoxyribonucleotides
- HU causes cell cycle arrest at or near the **G_1-S** interphase
- Therapeutic use

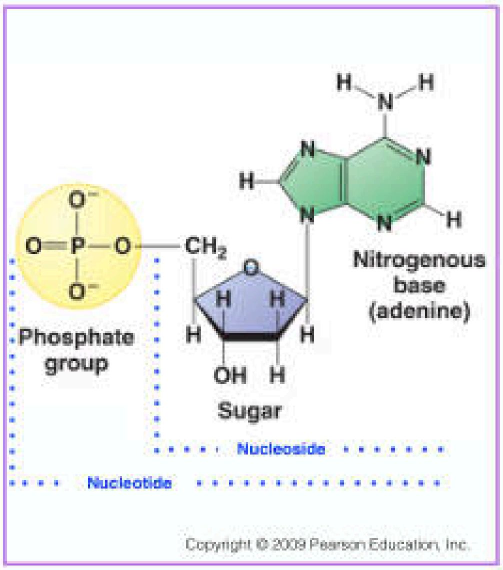
  : treat a certain type of chronic myelogenous leukemia (CML)



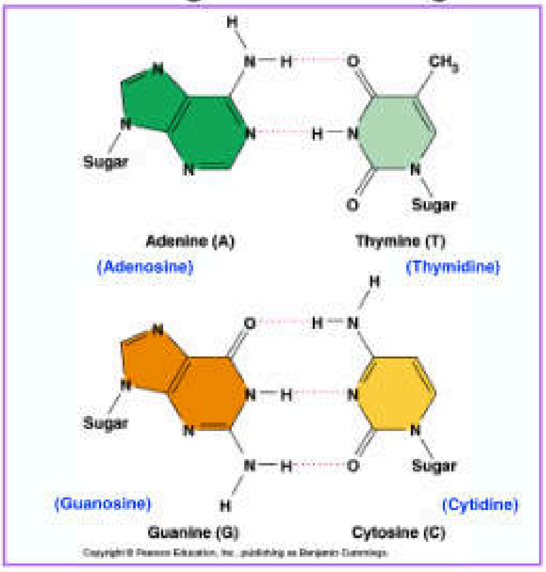
Purine & Pyrimidine Analogs



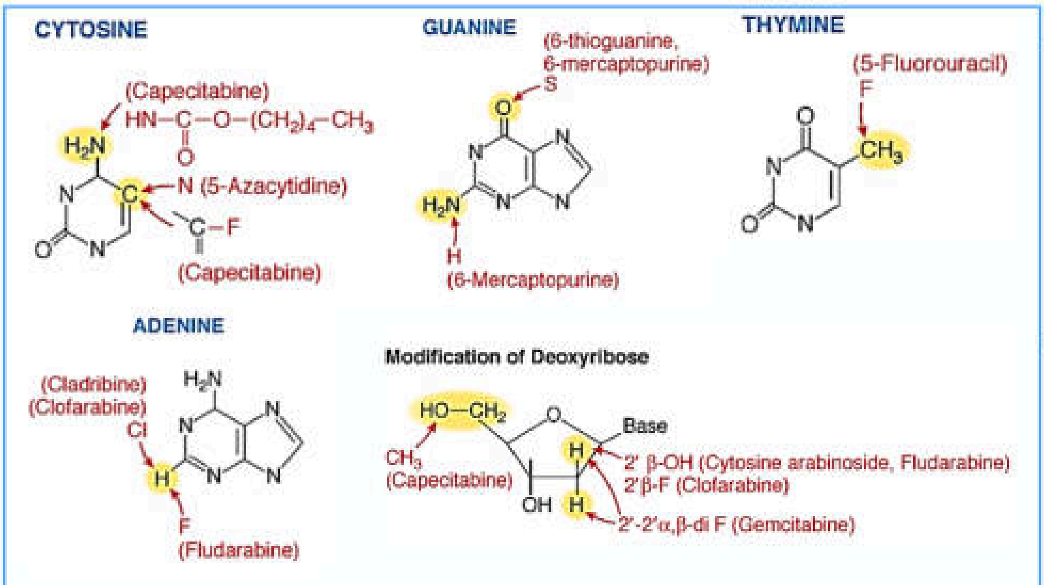
Structure of Nucleotide



Nitrogen Base Paring



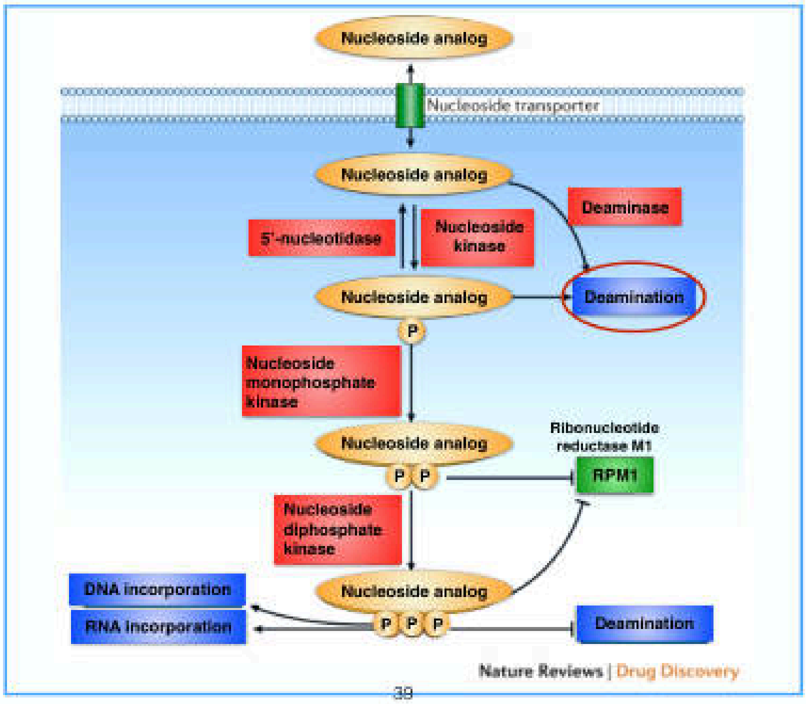
Structural Modification of Base and Deoxyribonucleoside Analogs



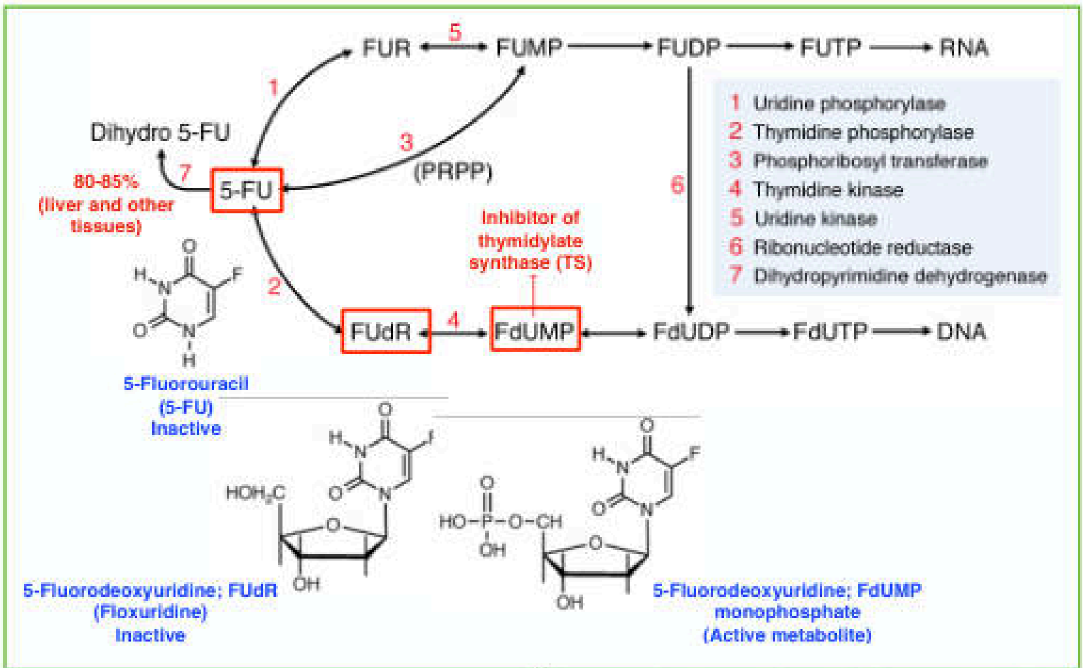
Antitumor analogs of purines and pyrimidines:
- *can be formulated as substituted bases (5-FU, 6-thioguanine)*
- *can be utilized as deoxynucleotides (Ara-C, gemcitabine, 5-azacytidine, cladribine)*

- **Pyrimidine analogs**
  - 5-Fluorouracil (5-FU)
  - Capecitabine
  - Cytarabine (Ara-C)
  - Gemcitabine (dFdC)
  - Azacitidine (5-Azacitidine)
- **Purine analogs**
  - 6-Thiopurines
  - Fludarabine phosphate
  - Cladribine
  - Clofarabine
  - Nelarabine
  - Pentostatin (2'-Deoxycoformycin)

Mechanism of action of nucleoside analogs

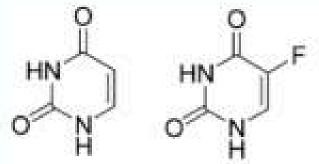


Fluoropyrimidine Analogs



Goodman and Gilman's: The Pharmacological Basis of Therapeutics, 12th Ed

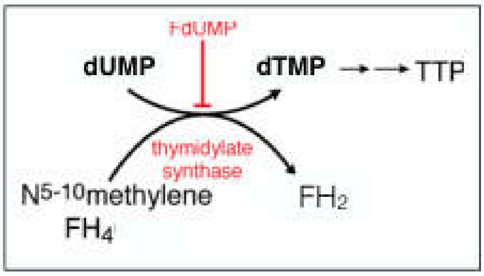
5-FU

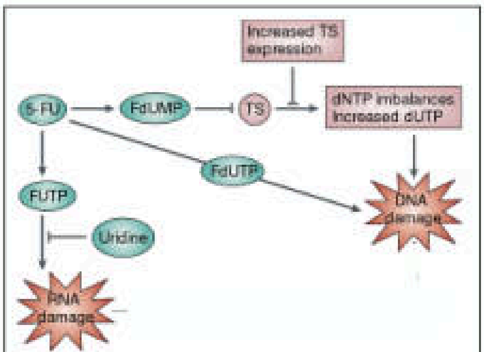


uracil 5-fluorouracil

- **IV**
  - ➠ absorption from GI tract is unpredictable and incomplete
- The most widely used agent in the treatment of colorectal cancer
- **Short** half-life (10-15 min)
- Metabolism
  - ➠ 80-85% of IV dose is catalyzed by **dihydropyrimidine dehydrogenase (DPD)** particularly in the liver, intestinal mucosa, tumor cells and other tissues
  - ➠ DPD deficiency is seen in 5% of cancer patients
- Toxicities: myelosuppression, mucositis, diarrhea, nausea, vomiting, **hand-foot syndrome**, and neurotoxicity

5-FU: Mechanisms





Nature Reviews Cancer 2003

- FdUMP inhibition of TS and block the synthesis of thymidine triphosphate (TTP)
- Incorporation of 5-FdUTP into DNA may result in DNA strand break
- Incorporation of 5-FUTP into RNA affecting the processing and function of RNA

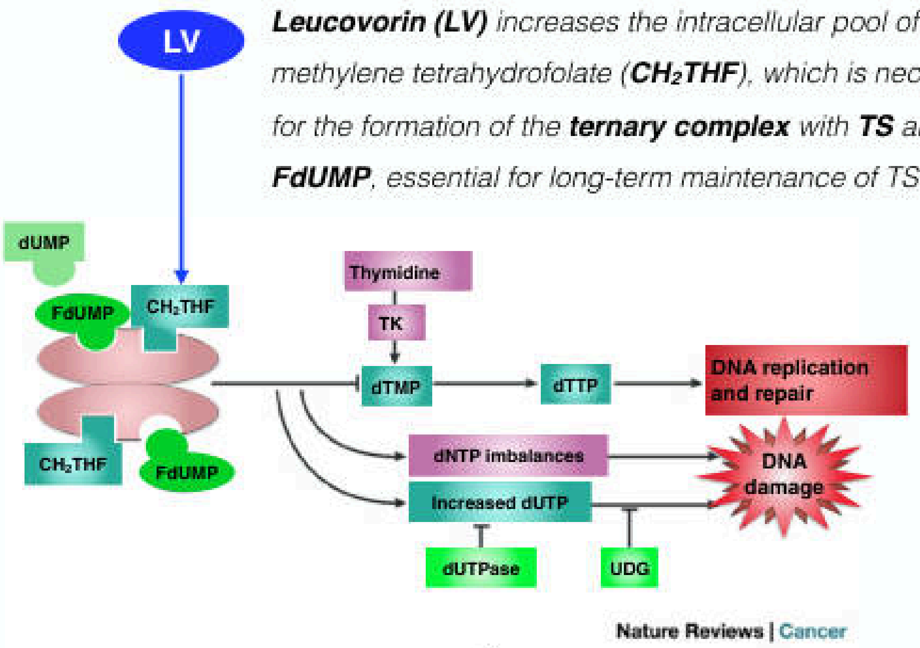
TS, Thymidylate synthase

FOLFOX regimen for the treatment of colorectal cancer

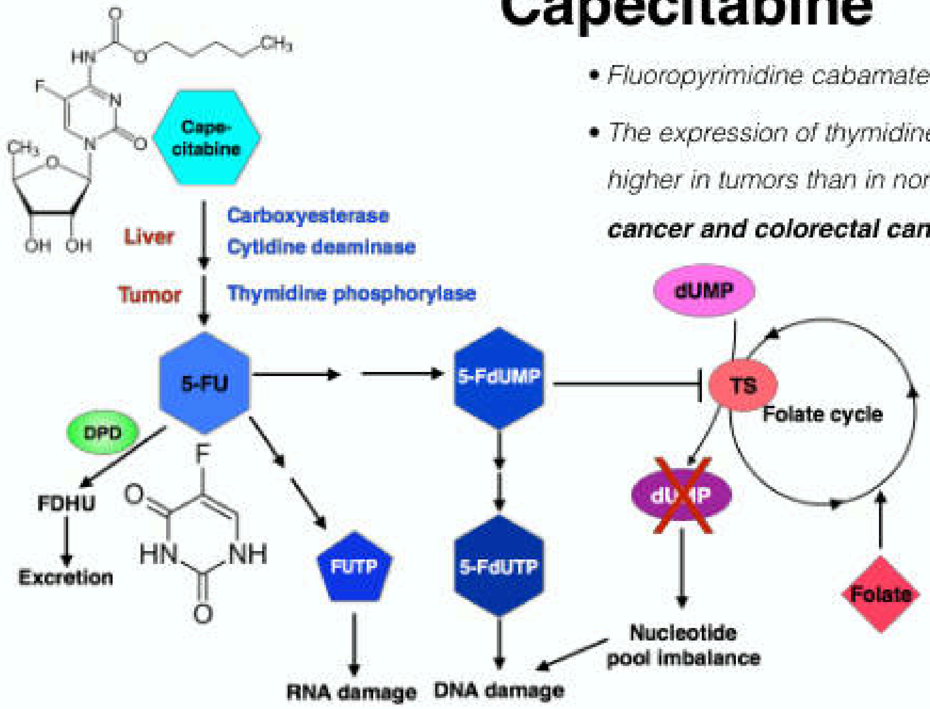
- *5-Fluorouracil*
- *Leucovorin* ? ? ?
- *Oxaliplatin*

Modulation of 5-FU Activity by Leucovorin

Leucovorin (LV) *increases the intracellular pool of 5,10-methylene tetrahydrofolate (**CH_2THF**), which is necessary for the formation of the **ternary complex** with **TS** and **FdUMP**, essential for long-term maintenance of TS inhibition*



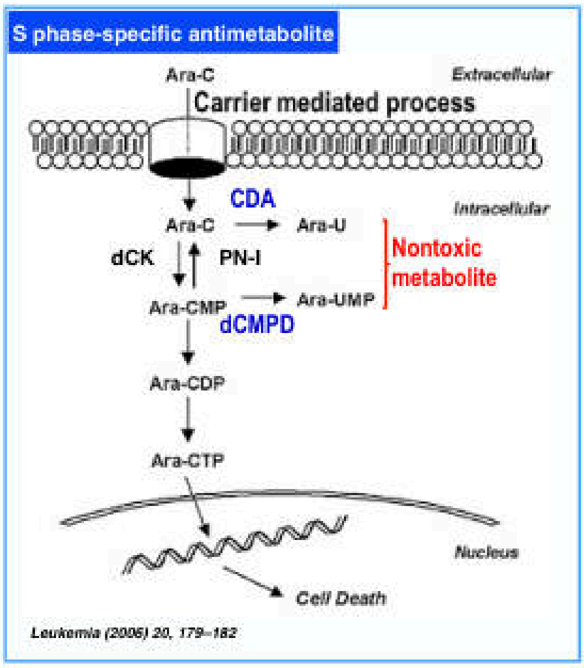
Capecitabine

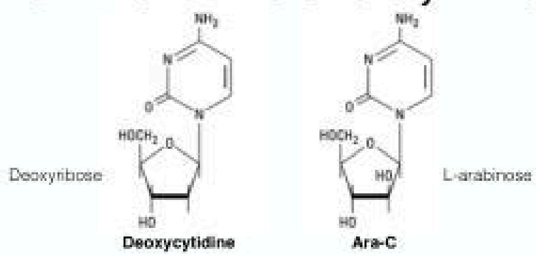


- *Fluoropyrimidine cabamate* ***prodrug***
- *The expression of thymidine phosphorylase is significantly higher in tumors than in normal tissue (particularly* ***breast cancer and colorectal cancer****)*

- *PO (bioavailability 70-80%)*
- *Main toxicities: diarrhea and the hand-foot syndrome*
- *Myelosuppression, nausea, vomiting, and mucositis are less than 5-FU*

Cytarabine (Cytosine Arabinoside; Ara-C)





- ❖ Incorporation of ara-CTP into DNA and RNA
- ❖ Ara-CTP competitively inhibits DNA polymerase-α and -β
  - ➠ Blockade of DNA synthesis
  - ➠ Blockade of DNA repair
- ❖ The cellular retention of ara-CTP appears to correlate with its lethality to malignant cells

dCK, deoxycytidine kinase; PN-I, pyrimidine nucleotides; CDA, cytidine deaminase; dCMPD, deoxycytidylate deaminase

Ara-C

- **Administration:**
  - Intravenous **infusion**
  - **Intrathecal**

➠ *clinical activity is highly schedule-dependent and rapid degradation by deamination*

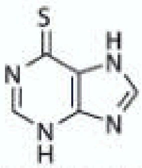
➠ *should **not** be administered **orally** due to high levels of **cytidine deaminase** in GI mucosa and liver (bioavailability ~20%)*

- Limited exclusively to hematologic malignancies

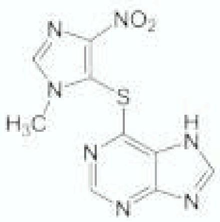
➠ *absolutely **no activity in solid tumors***

- **Adverse effects:**
  - Potent myelosuppression
  - Mucositis, nausea, vomiting, and neurotoxicity (high dose)

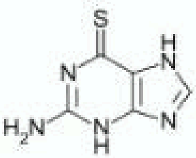
6-Thiopurines: Mechanism of Action



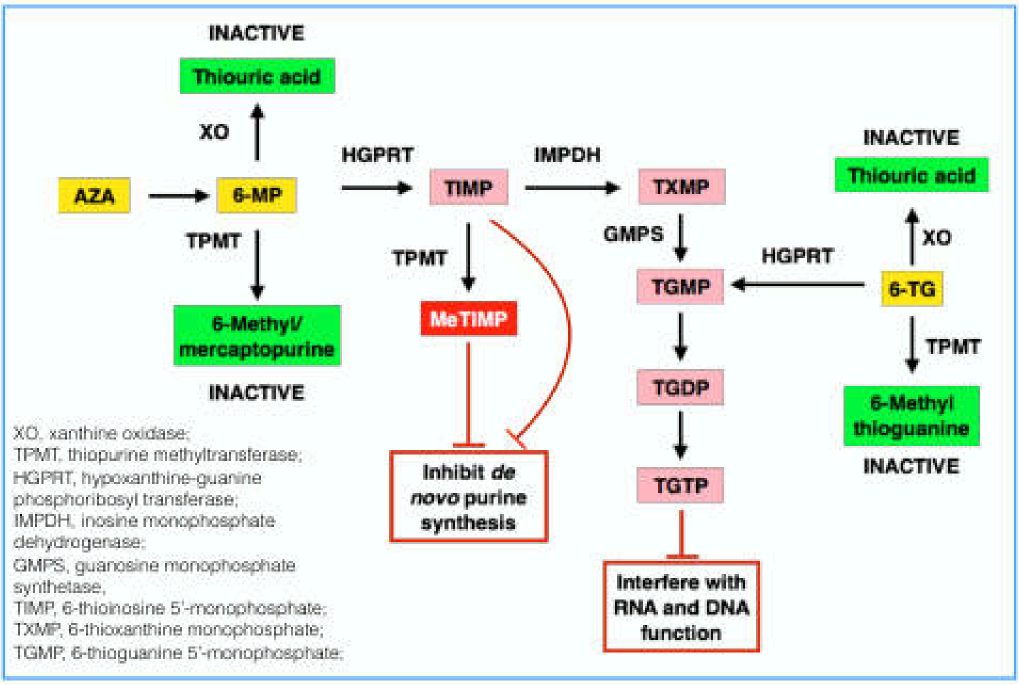
6-Mercaptopurine (6-MP)



Azathioprine (AZA)



6-Thioguanine (6-TG)



6-Mercaptopurine (6-MP)

Pharmacokinetics

- ✦ Absorption: 10-50% (oral ingestion)
- ✦ Metabolism: first-pass metabolism by xanthine oxidase in the **liver**
  - Converted to 6-thiouric acid (inactive metabolite) by **xanthine oxidase**
    - ➠ *Drug interaction with xanthine oxidase inhibitor (allopurinol)*
    - ➠ *Reduce oral dose of 6-MP by 50-75% (not required for IV dosing)*
  - Also metabolized by **thiopurine methyltransferase** (TPMT)
    - ➠ *15% of the Caucasian has decreased enzyme activity*
    - ➠ *Testing polymorphism in patients with auto-immune disease is recommended*

✦ **Toxicities**

- **Myelosuppression** (developing more gradual than folic analogs)
- **GI side effects**
  - ➠ Common: nausea, vomiting, and anorexia
  - ➠ Rare: stomatitis and diarrhea
  - ➠ Jaundice and ↑hepatic enzyme……..one-third of adult patients

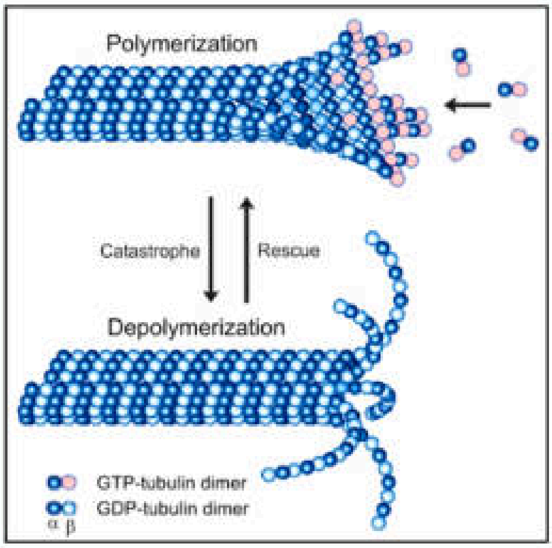
Antimetabolites	Mechanism of Action	Clinical Applications	Toxicity
Methotrexate	Inhibits DHFR and purine nucleotide synthesis	Breast cancer, head and neck cancer, osteogenic sarcoma, primary CNS lymphoma, non-Hodgkin's lymphoma, bladder cancer, choriocarcinoma	• Mucositis, diarrhea • Myelosuppression
5-Fluorouracil	Inhibits TS, alteration of RNA processing, inhibition of DNA synthesis and function	Colorectal cancer, anal cancer, breast cancer, gastroesophageal cancer, head and neck cancer, hepatocellular carcinoma	• Nausea and vomiting, diarrhea • Myelosuppression • Neurotoxicity
Capecitabine	Same as 5-FU	Breast cancer, colorectal cancer, gastroesophageal cancer, hepatocellular carcinoma, pancreatic cancer	• Nausea and vomiting, diarrhea • Hand-foot syndrome • Myelosuppression
Cytarabine	Inhibits DNA synthesis and repair, inhibits ribonucleotide reductase	Acute myeloid leukemia (AML), acute lymphoblastic leukemia (ALL), chronic myeloid leukemia (CML) in blast crisis	• Nausea and vomiting • Myelosuppression • Cerebellar ataxia
Gemcitabine	Same as cytarabine	Pancreatic cancer, bladder cancer, breast cancer, NSCLC, ovarian cancer, non-Hodgkin's lymphoma, soft tissue sarcoma	• Nausea and vomiting, diarrhea • Myelosuppression
6-Mercaptopurine (6-MP)	Inhibits de novo purine nucleotide synthesis, incorporation of triphosphate into RNA or DNA	AML	• Myelosuppression • Immunosuppression • Hepatotoxicity
6-Thioguanine (6-TG)	Same as 6-MP	ALL, AML	Same as 6-MP

Basic and Clinical Pharmacology, *12th Ed*

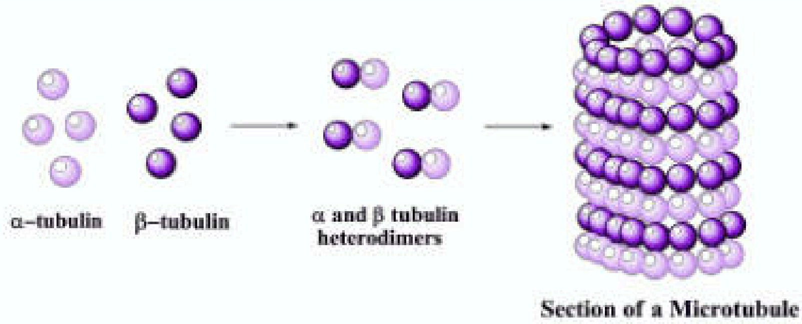
Natural Product Cancer Chemotherapeutic Drugs

- Microtubule-Targeting Agents
- Topoisomerase Inhibitors

Microtubule Targeting Agents

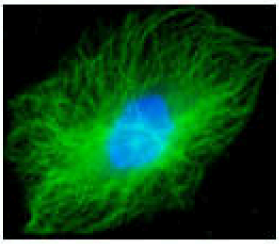


Microtubule

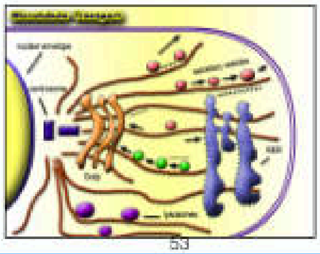


Functions

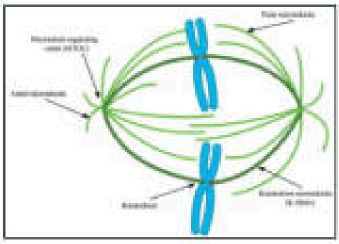
Filamentous intracellular structure



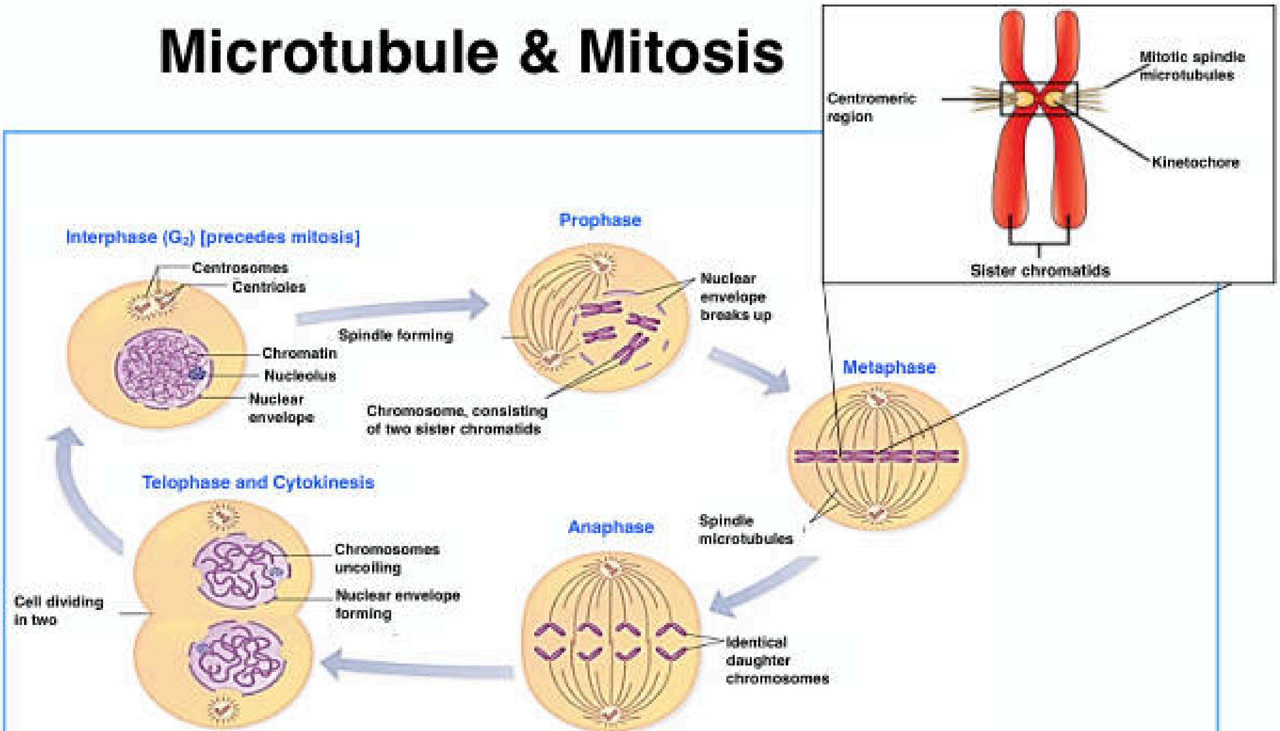
Cellular transport



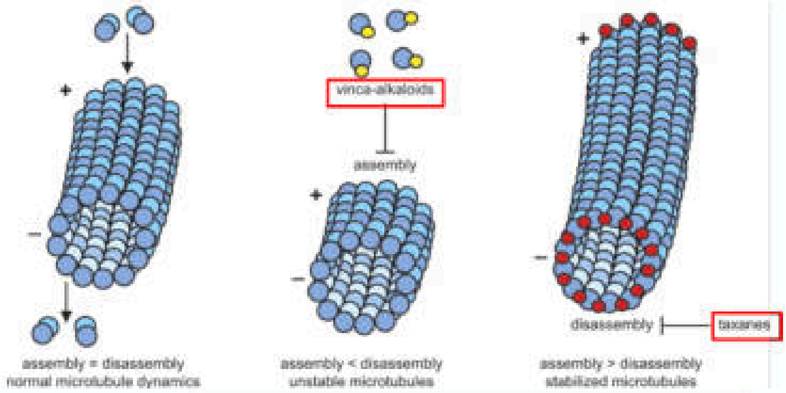
Cell division



Microtubule & Mitosis



Microtubule-Targeting Agents (MTAs)

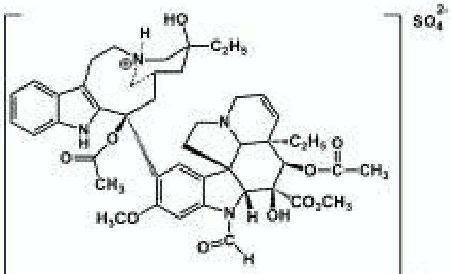


MTAs

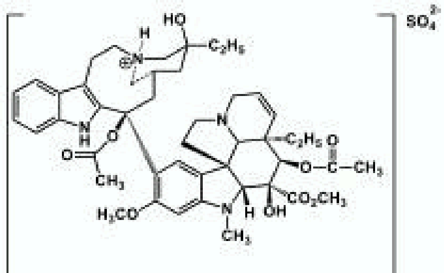
- **Microtubule-destabilizing agents**
  - ~ **Vinca alkaloids**
- **Microtubule-stabilizing agents**
  - ~ **taxanes, epothilones**

Genes Dev. 2006 Aug 1;20(15):1975-81.

Vinca Alkaloids

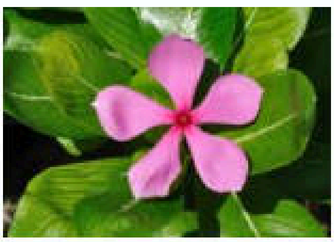


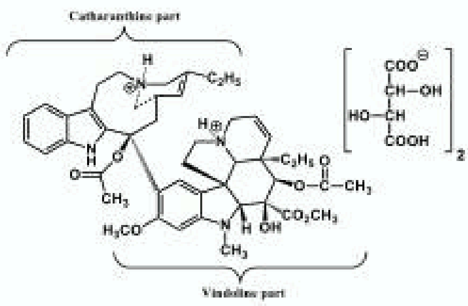
Vincristine Sulfate (Vincasar PFS)



Vinblastine Sulfate (Velban)

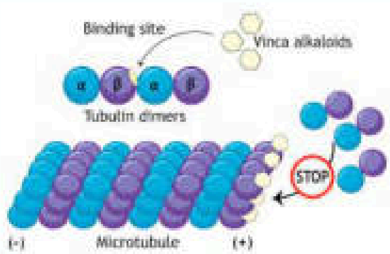
- Alkaloids derived from the periwinkle plant (*Catharanthus rosea*)





Vinorelbine tartrate (Navelbine)

Vinca Alkaloids



Mechanism of action

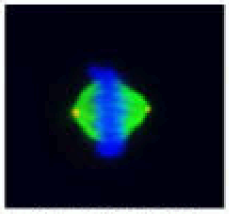
- Cell-cycle specific agents
- Bind specifically to β tubulin ➠ block microtubule **polymerization** ➠ block cells in mitosis

Pharmacokinetics

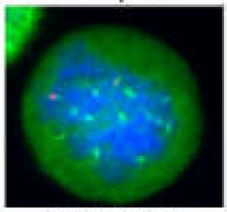
- Metabolized by the liver P450 system (mainly **CYP3A**)
- <15% of a dose is found in the urine unchanged
- ↓dose 50-75% in patient with hepatic dysfunction (bilirubin > 3 mg/dL)

Toxicities

- Myelosuppression (very limited with vincristine)
- Neurotoxicity (rare and mild with vinblastine)
- GI disturbance: nausea, vomiting, anorexia, and diarrhea
- Alopecia



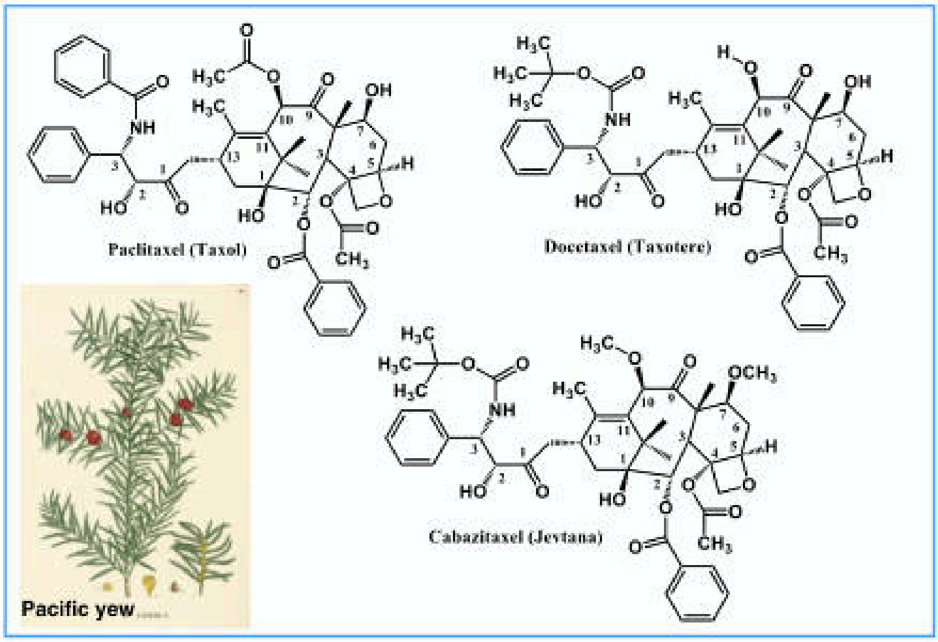
Normal microtubule at Metaphase



Depolymerized microtubules

Fatal misadministration of intrathecal Vinca alkaloids

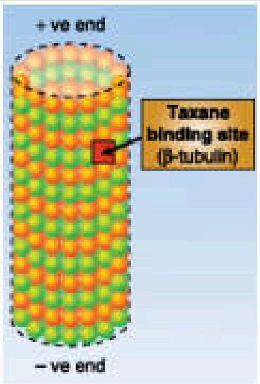
Taxanes



Taxanes

Mechanism of action

- Cell-cycle specific agents
- Bind specifically to β tubulin
  - ➠ **block** microtubule **depolymerization**
  - ➠ block cells in mitosis

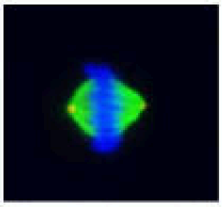


Pharmacokinetics

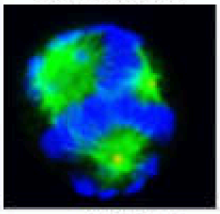
- Metabolized: paclitaxel ➠ **CYP2C8** and CYP3A4; docetaxel ➠ **CYP3A4 and CYP3A5**
- <10% of a dose is excreted in the urine intact
- ↓dose 50-75% in patient with hepatic dysfunction
- Clearance markedly delayed by cyclosporine (P-glycoprotien inhibitor)

Toxicities

- **Myelosuppression** (docetaxel causes neutropenia > paclitaxel)
- Peripheral neuropathy (dose limiting toxicity)…(paclitaxel > docetaxel)
- Hypersensity reactions (5%)
  - ➠ **Pretreatment** with dexamethasone, diphenhydramine, and H_2 blocker
  - ➠ Albumin-bound paclitaxel formulation is not associated with hypersensitivity reaction

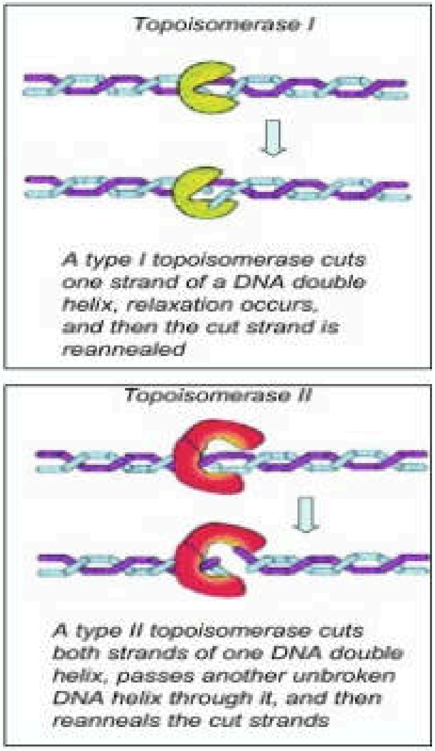


Normal microtubule at Metaphase



Stabilized microtubules

Topoisomerase Inhibitors

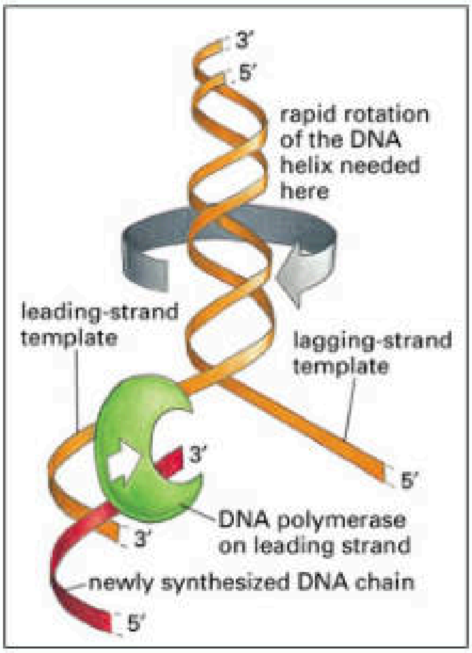


Studies in Natural Products Chemistry, Volume 54, 2017, Pages 21-86.

Topoisomerases

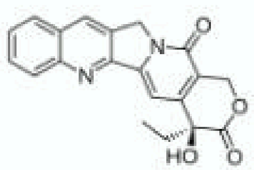
Function of Topoisomerases

- Regulation of the overwinding or underwinding of DNA during DNA replication
- **Topoisomerase I**: Relax DNA by Nicking and Then Closing One Strand of Duplex DNA
- **Topoisomerase II**: Change DNA topology by breaking and Rejoining Double-Stranded DNA

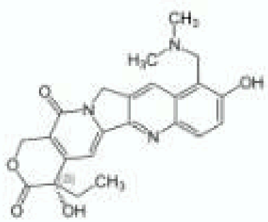


Molecular Biology of The Cell. 4th Ed.

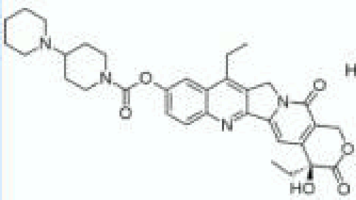
Topoisomerase I Inhibitors: Mechanism of Action



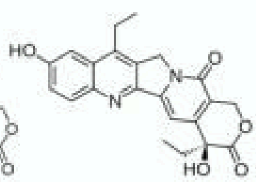
Camptothecin (CPT)



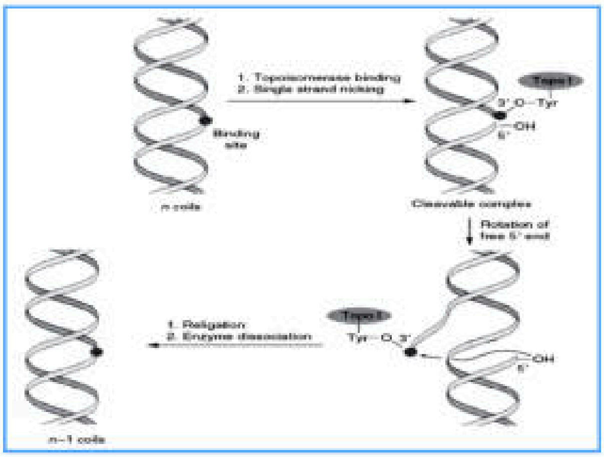
Topotecan

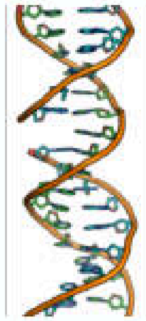


Irinotecan (CPT-11)



SN-38





Mechanism of action

- Bind to and stabilize DNA-topoisomerase I complex ➠ **DNA damage**

Topoisomerase I Inhibitors

Topotecan

✦ **Pharmacokinetics**

- 30-40% of the administered dose in urine within 24 h

✦ **Toxicities**

- Myelosuppression (neutropenia ➠ dose-limiting toxicity)
- GI side effects: mucositis and diarrhea
- Other (less common and mild toxicities): nausea and vomiting, fever, rash, elevated liver transaminases

Irinotecan

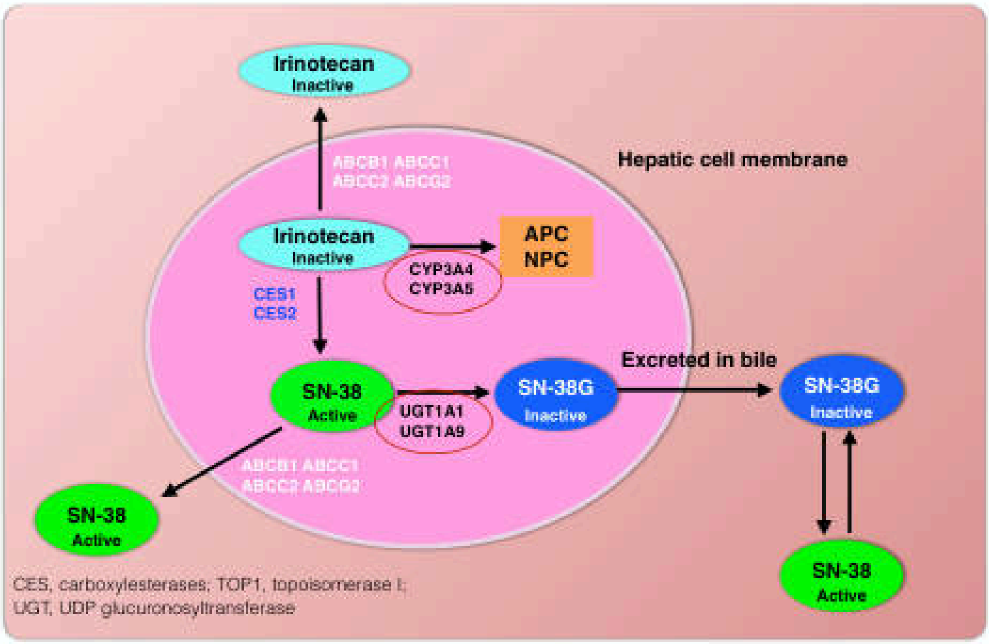
✦ **Pharmacokinetics**

- hepatic metabolism (CYP3A) is the major route of elimination

✦ **Toxicities**

- **Cholinergic syndrome** (within 24 h): acute diarrhea, diaphoresis, hypersalivation, etc
- Delayed diarrhea (dose-limiting toxicity).....Tx: loperamide
- Myelosuppression

Metabolism of Irinotecan

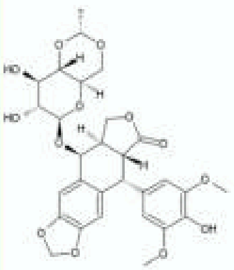


SN-38 ~1000-fold more potent than irinotecan

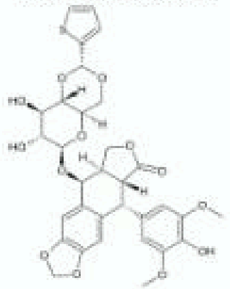
Topoisomerase II Inhibitors

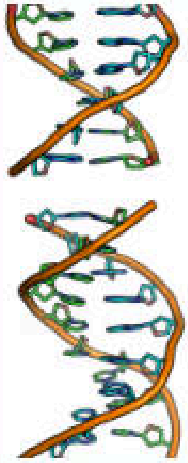
- Semisynthetic derivatives of podophyllotoxin, which is extracted from the mayapple root
- Having significant therapeutic activity in pediatric leukemia, small cell lung cancer (SCLC), testicular tumors, Hodgkin's lymphoma, and large cell lymphomas

Etoposide (VP-16-213)



Teniposide (VM-26)

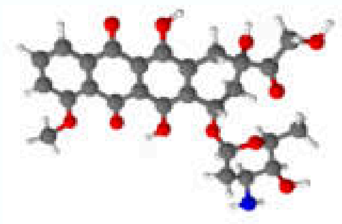




- Form ternary complex with DNA and topoisomerase II ➠ **DNA damage**
- Cells in S and G_2 phases are more sensitive
- ~40% of drug is excreted in the urine ➠ ↓dose in renal dysfunction
- Dose-limiting toxicity is leukopenia

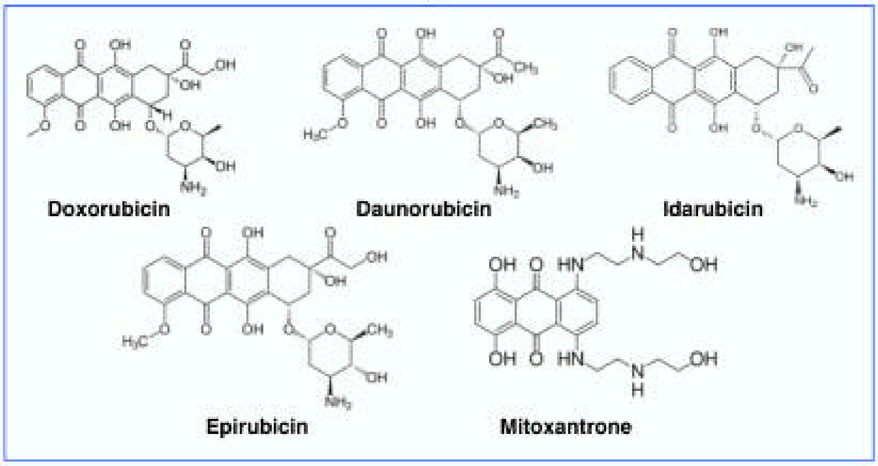
Antimetabolites	Mechanism of Action	Clinical Applications	Toxicity
Vinblastine	Inhibits mitosis	Hodgkin's and non-Hodgkin's lymphoma, germ cell cancer, breast cancer, Kaposi's sarcoma	• Nausea and vomiting • Myelosuppression • Mucositis, alopecia, SIADH
Vincristine	Inhibits mitosis	ALL, Hodgkin's and non-Hodgkin's lymphoma, rhabdomyosarcoma, neuroblastoma, Wilms' tumor	• Peripheral neuropathy • Paralytic ileus • Myelosuppression • Alopecia, SIADH
Vinorelbine	Inhibit mitosis	NSCLC, breast cancer, ovarian cancer	• Nausea and vomiting • Myelosuppression • Constipation, SIADH
Paclitaxel	Inhibits mitosis	Breast cancer, NSCLC, SCLC, ovarian cancer, gastroesophageal cancer, prostate cancer, bladder cancer, head and neck cancer	• Nausea and vomiting • Hypersensitivity • Myelosuppression • Peripheral neuropathy
Docetaxel	Inhibits mitosis	Breast cancer, NSCLC, prostate cancer, gastric cancer, head and neck cancer, ovarian cancer, bladder cancer	• Hypersensitivity • Neurotoxicity • Myelosuppression
Topotecan	Inhibits topoisomerase I	SCLC, ovarian cancer	• Nausea and vomiting • Myelosuppression
Irinotecan	Inhibits topoisomerase I	Colorectal cancer, NSCLC, SCLC, gastroesophageal cancer	• Diarrhea, nausea and vomiting • Myelosuppression
Etoposide	Inhibit topoisomerase II	NSCLC, non-Hodgkin's lymphoma, gastric cancer	• Nausea and vomiting • Myelosuppression • Alopecia

Antitumor Antibiotics



- Anthracyclines
- Mitomycin
- Bleomycin
- Neocarzinostatin

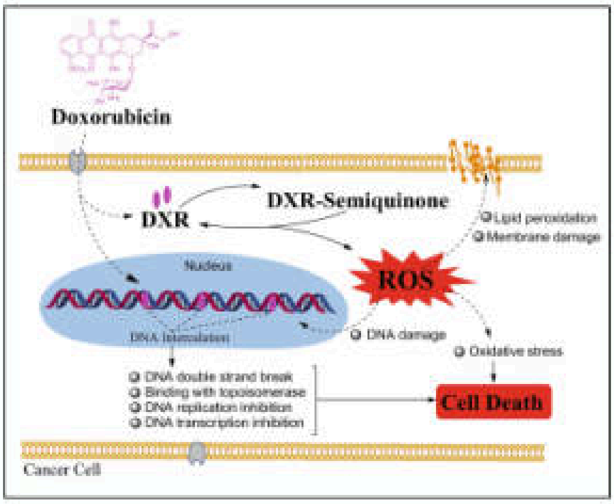
Anthracyclines



- Anthracyclines are derived from *Streptomyces peucetius* var. *caesius*
- Idarubicin and epirubicin are analogs of doxorubicin and daunarubicin
- They posses potential for generating **free radicals**
- One of the most widely used cytotoxic anticancer drugs

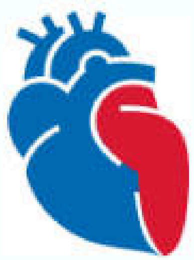
Anthracyclines: Mechanism of Action

- Inhibition of **topoisomerase II**
- High-affinity binding to DNA through **intercalation**
  - ➠ *Blockade of the synthesis of DNA and RNA*
  - ➠ *DNA strand scission*
- Generation of semiquinone free radicals and oxygen **free radicals**
- Binding to cellular membranes to alter **fluidity** and ion transport



Pharmaceuticals from Microbes. Environmental Chemistry for a Sustainable World, vol 26. 2019

Anthracyclines: Toxicities



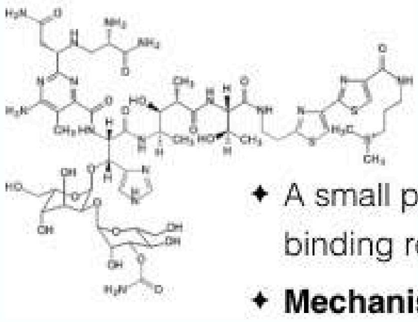
- Main dose-limiting toxicity of all anthracyclines is **myelosuppression**
- **Two forms of cardiotoxicity are observed**

  : Acute toxicity ➠ arrhythmias and conduction abnormalities

  : Chronic toxicity ➠ **cumulative** dose-dependent, dilated cardiomyopathy

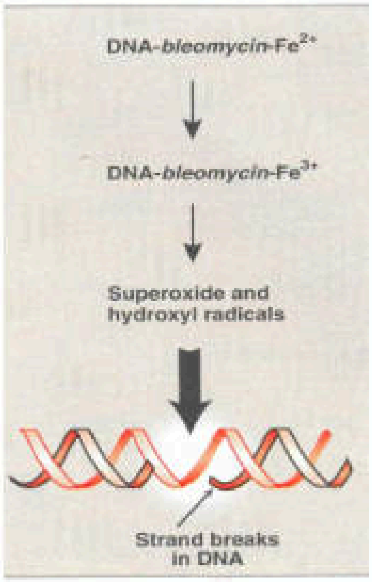
  (usually total dose for adult ≥550 mg/m²)

(Dilated cardiomyopathy, heart becomes weakened and enlarged and cannot pump blood efficiently)

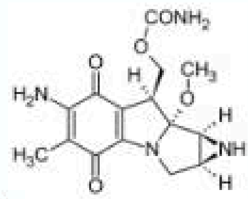


Bleomycin

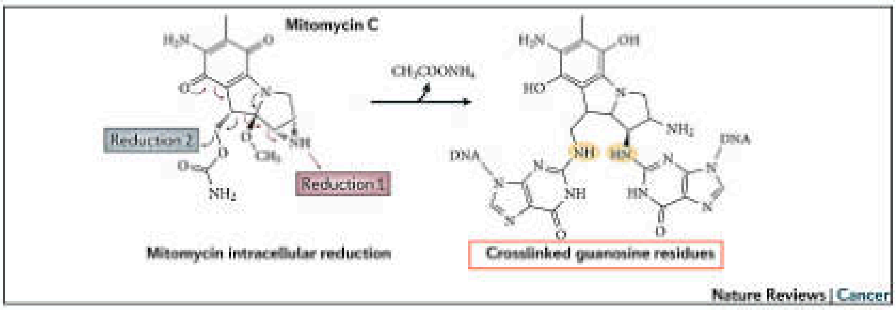
- A small peptide that contains a DNA-binding region and an iron-binding domain
- **Mechanism of action**
  - ➠ *binding to DNA*
  - ➠ *radical formation*
  - ➠ *resulting in single- and double-strand breaks*
- Cell cycle specific drug that causes accumulation of cells in the **G_2 phase**
- **Pulmonary toxicity** is dose-limitation with increasing incidence in patients >70 years



Mitomycin C



- Isolated from *Streptomyces caespitosus*
- **Mechanism of action**

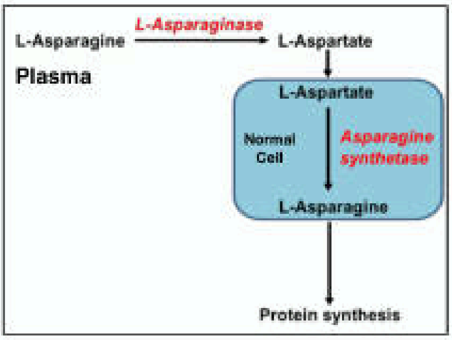


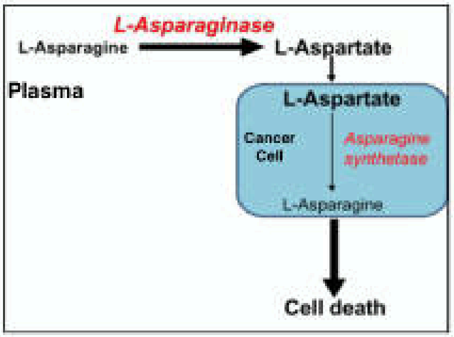
- **Hypoxic tumor** stem cells of solid tumors exist in an environment conducive to reductive reactions that are **more sensitive** to the cytotoxicity of mitomycin than normal cells and oxygenated tumor cell

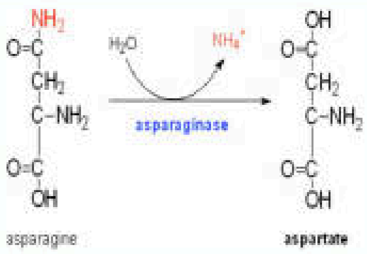
Antimetabolites	Mechanism of Action	Clinical Applications	Toxicity
Doxorubicin	• Oxygen free radicals • Inhibits topoisomerase II • Intercalates into DNA	Breast cancer, Hodgkin's and non-Hodgkin's lymphoma, soft tissue sarcoma, ovarian cancer, NSCLC, SCLC, thyroid cancer, Wilms' tumor, neuroblastoma	• Nausea • Red urine (not hematuria) • Cardiotoxicity • Myelosuppression • Alopecia • Mucositis
Daunarubicin	Same as doxorubicin	AML, ALL	
Idarubicin	Same as doxorubicin	AML, ALL, CML in blast crisis	
Mitomycin	• Acts as alkylating agent and form cross-links with DNA • Formation of oxygen free radicals	Superficial bladder cancer, gastric cancer, breast cancer, NSCLC, head and neck cancer (in combination with radiotherapy)	• Nausea and vomiting • Myelosuppression • Mucositis • Anorexia and fatigue • Hemolytic uremic syndrome
Bleomycin	• Formation of oxygen free radicals • Single- and double-strand DNA breaks	Hodgkin's and non-Hodgkin's lymphoma, germ cell cancer, head and neck cancer	• Allergic reactions • Fever • Hypotension • Skin toxicity • Pulmonary fibrosis • Mucositis • Alopecia

Miscellaneous Agent: **L-Asparaginase**

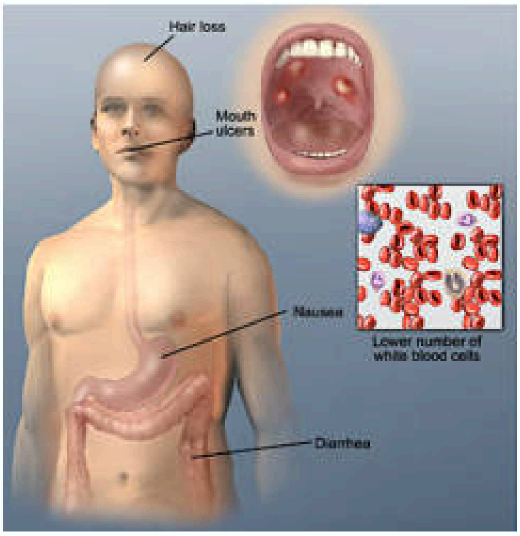
- Most normal tissues are able to synthesize L-asparagine in amount sufficient for protein synthesis
- Lymphocytic leukemia lack adequate amount of asparagine synthetase
- L-Asparaginase becomes a standard agent for treating acute lymphocytic leukemia (ALL)







Adverse Effects of Chemotherapy



http://www.mesothelioma.com/treatment/conventional/chemotherapy/

The toxicities associated with chemotherapy are the most important factors limiting the use of potentially curative doses

Common and Acute Toxicities

Organs with active cell division are affected

- Bone marrow suppression
  - ➠ *Most myelosuppressive regimens recommend dosing intervals of 3 to 4 wks*
- GI tract mucosa
  - ➠ *Nausea and vomiting*
  - ➠ *Stomatitis and xerostomia (dry mouth)*
  - ➠ *Esophagitis*
  - ➠ *Lower GI tract complications: malabsorption, diarrhea, or constipation*
- Dermatologic toxicities
  - ➠ *Alopecia*
  - ➠ *Dry skin*
  - ➠ *Hand-foot syndrome*

Specific Organ Toxicities

✦ **Neurotoxicity**: cytarabine and L-asparaginase

✦ **Cardiac toxicity**: doxorubicin, daunorubicin, idarubicin

✦ **Nephrotoxicity**: cisplatin

✦ **Pulmonary toxicity**: bleomycin

✦ **Hepatotoxicity**: L-asparaginase, imatinib, carmustine, methotrexate

Long-Term Complications

✤ **Secondary malignancies**

➠ *Unrelated to the first cancer that was treated*

➠ *Normal cells that are especially sensitive to chemotherapy and most likely to begin carcinogenesis (the bone marrow, hair follicles, and the epithelial cells of the GI tract)*

➠ *May occur months or even years after initial treatment*

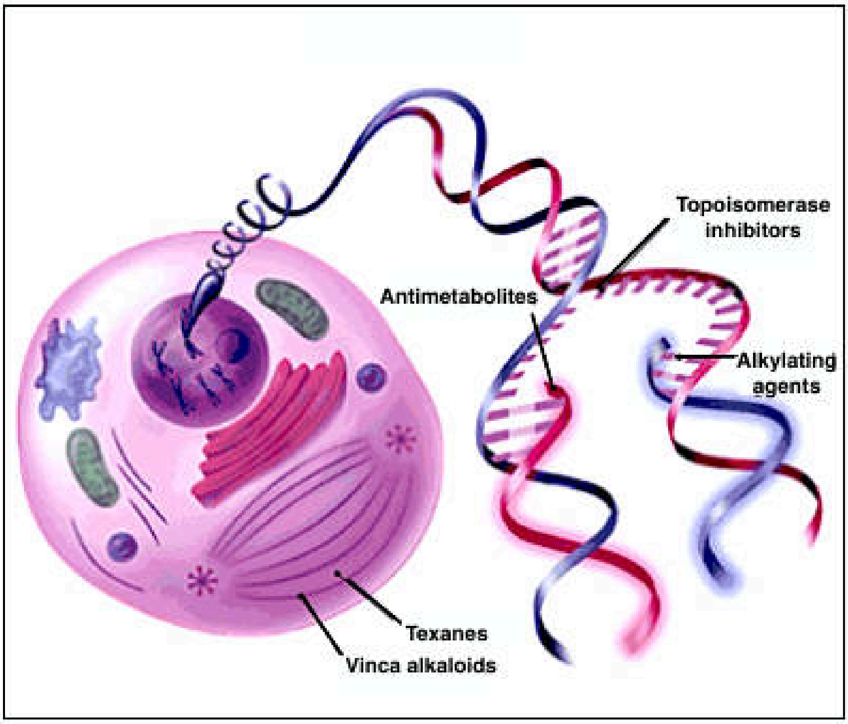
➠ The development of secondary hematologic cancers such as leukemia and lymphoma pose the greatest risk to adult and childhood cancer survivors

✤ **Fertility and teratogenicity**

➠ *Chemotherapy is potentially gonadotoxic in human*

➠ *The risk of producing an abnormal offspring would be highest at the time of ongoing germ cell exposure*

Mechanism of Chemotherapeutic Agents



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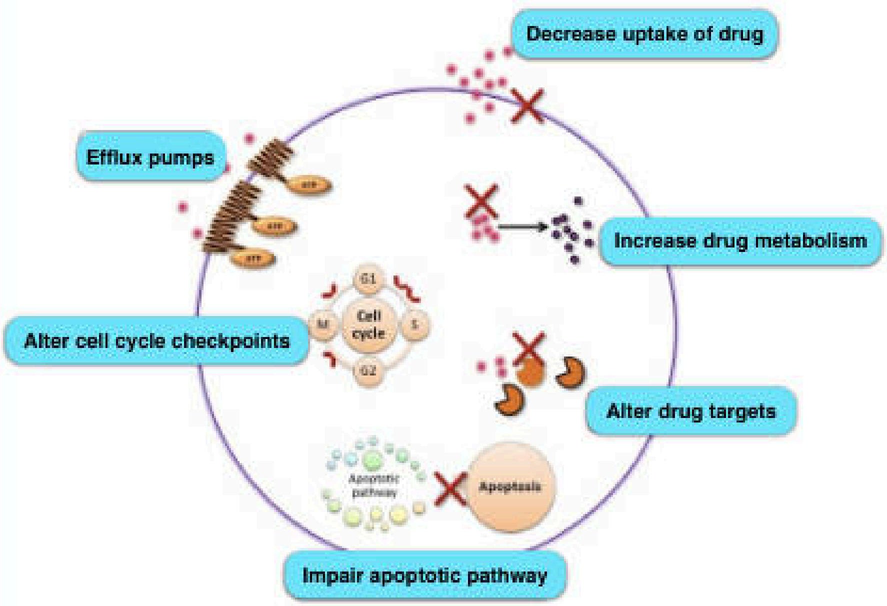
Am Fam Physician. 2008 Feb 1;77(3):311-319

Mechanisms of Chemotherapeutic Resistance



Chai et al. Chinese Medicine 2010 5:26.

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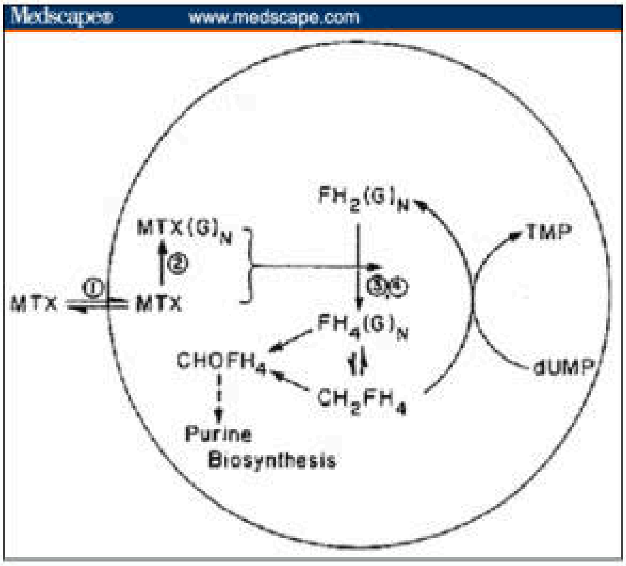
Example: Resistance to Methotrexate



- ⬇ drug transport (⬇ reduced folate carrier or folate receptor protein)
- ⬇ formation of toxic MTX polyglutamates
- ⬆ levels of the target enzyme DHFR
- Altered DHFR protein with reduced affinity for MTX